# AMA Facial Recognition: Digital Record-logs Management System

**A Thesis Presented to the Faculty of AMA Computer**

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Degree Bachelor of Science in Computer Science

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AMA COMPUTER COLLEGE
LUCENA CITY, INC.

**APPROVAL SHEET**

This research proposal entitled "**AMA FACIAL ECOGNITION: DIGITAL RECORD-LOGS MANAGEMENT SYSTEM**" prepared and submitted by **JOHN EMMANUEL S. MARCO** and **JOHN KENNEDY H. PEÑA** in partial fulfilment of the course requirements for the degree of **Bachelor of Science in Computer Science** has been examined and recommended for acceptance and approval for Oral examination.

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## ABSTRACT

This study presents the development of the AMA Facial Recognition: Digital Record-logs Management System, a local network-based solution designed to enhance campus security and streamline administrative operations at AMA Computer College. By integrating facial recognition technology with existing CCTV infrastructure, the system automates the identification and logging of individuals in real-time, significantly reducing manual workload and minimizing the risk of human error in record management. Data is securely stored using a multi-format approach: facial recognition details in XML, general records in a JSON-based database, and activity logs in CSV files. These files are accessible only within the college's local network, ensuring data privacy and protection. The system not only improves data accuracy and operational efficiency but also empowers administrators with tools for timely monitoring, reporting, and response. This project demonstrates a forward-thinking application of biometric technology in education, offering a scalable, secure, and efficient solution for modern record-keeping challenges.

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# CHAPTER I
# PROJECT AND ITS BACKGROUND

## Project Background

In the digital age, managing records and logs manually can be fraught with inefficiencies and risks. Traditional methods of record-keeping, which rely heavily on manual paperwork, making it difficult to streamline operations, ensure data accuracy, and maintain security. Manual processes often lead to errors in data entry, misfiling of documents, and delays in information retrieval, which can significantly hamper productivity.

The implementation of facial recognition technology presents a promising solution to these challenges. By leveraging facial recognition for digital record management, organizations can significantly enhance the accuracy and security of their records. Facial recognition systems use sophisticated algorithms to match a person's facial features with stored data, ensuring that only authorized individuals can access sensitive information. This not only enhances security but also improves the efficiency of the record-keeping process by eliminating the need for manual verification.

## Project Context

Inefficient record-keeping is a significant challenge at AMA Computer College, where managing records manually leads to time-consuming processes and a high likelihood of errors. Communicating and managing records about student attendance, staff presence, or visitor logs becomes increasingly difficult without a centralized digital platform. This often results in miscommunication or delays in accessing crucial information, affecting

the overall operational efficiency of the institution.

Manual record management poses accessibility challenges, making it inconvenient for administrators to retrieve and verify records quickly. This inefficiency not only hampers administrative productivity but also increases the risk of data loss or misplacement of important documents. Additionally, the absence of automated tracking systems makes it difficult to monitor and verify the presence and activities of individuals on campus, which can compromise security and accountability.

The implementation of the AMA Facial Recognition: Digital Record-logs Management System aims to address these challenges by leveraging advanced facial recognition technology to automate the identification and logging process. This system will ensure accurate and secure digital records of all individuals passing in front of CCTV cameras on campus. By transitioning from manual to automated processes, the system will reduce administrative burdens, minimize errors, and enhance the overall security of record management. Furthermore, the digital platform will facilitate real-time updates and easy access to information, thereby improving the efficiency and effectiveness of the college's administrative operations.

**Purpose and Description**

The AMA Facial Recognition: Digital Record-logs Management System is an innovative solution designed to enhance campus security and streamline record-keeping at AMA Computer College. This system leverages advanced facial recognition technology integrated with existing CCTV infrastructure to automatically log and manage records of individuals passing through monitored areas. The core functionality revolves around processing video feeds from CCTV cameras in real-time, utilizing sophisticated facial recognition algorithms to identify individuals and create timestamped digital records of their presence on campus. Crucially, this system operates exclusively within the local

network of AMA College, ensuring all sensitive data remains within the institution's secure digital perimeter. The system's primary output is an Excel file, which serves as the central repository for all collected data, offering a familiar and accessible format for data storage and management.

This system is a multi-faceted data management approach that optimizes storage and accessibility. The system utilizes a JSON-based database to store comprehensive data, providing flexibility and efficiency in data handling. Facial recognition data is stored in XML files, allowing for structured storage of complex facial features and metadata. For logging purposes, the system generates CSV files, offering a simple and widely compatible format for recording system activities and events. Authorized personnel can access and manipulate these files only when connected to the college's local network, significantly enhancing data security by preventing external access. This multi-format approach allows for easy searching, sorting, and analysis of the digital record-logs, enabling staff to quickly retrieve information as needed. Real-time monitoring capabilities are integrated into the system, with the ability to update the JSON database, XML files, and CSV logs instantly as new data is captured. This feature provides immediate insights into campus activity, allowing for prompt responses to any security concerns or anomalies detected by the facial recognition system.

Security and privacy considerations are paramount in the design of the AMA Facial Recognition: Digital Record-logs Management System. The use of a JSON-based database for comprehensive data storage, XML files for facial recognition data, and CSV files for logs, combined with the system's confinement to the local network, adds multiple layers of security, significantly reducing the risk of unauthorized access or data breaches. Access to these files is strictly controlled, with robust encryption and authentication mechanisms ensuring that only authorized personnel can view or modify the data. By automating the process of detecting and logging individuals on campus into this multi-format data storage system, this solution not only enhances security measures but also significantly reduces the administrative burden on staff. The structured nature of JSON,

XML, and CSV formats ensures that the system remains efficient and easily maintainable, while still providing powerful data management capabilities. The JSON database offers flexibility for complex data structures, XML provides a standardized format for facial recognition data, and CSV logs offer straightforward activity tracking. Ultimately, the AMA Facial Recognition: Digital Record-logs Management System represents a significant step forward in leveraging technology to create a safer, more efficient educational environment, all while maintaining strict control over data access and security through its local network implementation and multi-format data storage solution.

**Objective of the Study**

The main objective of this research is to develop a facial recognition-based digital record-logs management system that revolutionizes campus security and operations at AMA Computer College. This system aims to offer enhanced security, efficiency, and better engagement.

Specifically, this study aims to:

1. Develop a system that efficiently organizes data and tracks individuals passing through CCTV-monitored areas using facial recognition software.
2. Create a comprehensive system capable of producing all necessary reports for security and administrative purposes.
3. Implement a secure, local network-based solution that stores facial recognition data in XML format, general data in a JSON-based database, and activity logs in CSV files.

**Significance of the Study**

The AMA Facial Recognition: Digital Record-logs Management System represents a

significant advancement in campus security and operational efficiency at AMA Computer College. This innovative system plays a pivotal role in enhancing day-to-day operations by automating the process of tracking individuals on campus through CCTV cameras. By streamlining security procedures and reducing manual workload, it ensures smooth operational efficiency while prioritizing campus safety.

The system's real-time monitoring and alert capabilities enable quick responses to security concerns, fostering a secure environment for students and staff. Moreover, its robust data management approach, utilizing JSON, XML, and CSV formats within a confined local network, demonstrates a strong commitment to data security and privacy protection. The implementation of this facial recognition system brings numerous benefits to administrative efficiency and technological integration.

It aids in various administrative tasks, from tracking campus population flow to generating comprehensive reports, thereby improving overall institutional management. By incorporating cutting-edge facial recognition technology into existing CCTV systems, AMA Computer College positions itself at the forefront of campus security innovations. This project not only enhances the college's technological infrastructure but also provides valuable opportunities for staff training and development in advanced security technologies. The successful implementation of this system, while maintaining privacy and ethical standards, can significantly boost community trust, showcasing the institution's dedication to safety and responsible technology use.

# PROJECT SCOPE AND LIMITATIONS

## Scope of the Study

The project will develop a facial recognition system integrated with existing CCTV cameras on the AMA Computer College campus. This system will capture and process facial data of individuals passing through monitored areas, providing a comprehensive solution for campus security and management. To ensure efficient data handling and storage, the project will implement a multi-faceted approach: a JSON-based database will be used for general data storage, XML files will store facial recognition data, and CSV files will be utilized for activity logs. Importantly, to maintain the highest level of security and data protection, the entire system will operate exclusively within the college's local network, preventing any unauthorized external access.

## Delimitations of the Study

The facial recognition system's functionality is constrained to areas monitored by existing CCTV cameras on the AMA Computer College campus. Its accuracy may vary depending on environmental factors such as lighting conditions, camera angles, and potential obstructions. Due to security considerations, the system is deliberately designed without off-campus or remote access capabilities. The system's performance and capacity are bound by the processing power and storage resources allocated by the college's IT infrastructure.

# CHAPTER II
# PROJECT AND ITS BACKGROUND

## Foreign Literature

In recent years, a number of face recognition based attendance management system have introduced in order to improve the performance of students in different organization. The human face is a sophisticated multidimensional structure that can convey a lot of information about the individual, including expression, feeling, facial features.

Face recognition is a technology created by Woodrow Wilson Bleadsoe in 1966 that works to match human faces through digital images or video footage through a facial database. Face recognition became an idea to allow computers to find and recognize human faces quickly and precisely. Many algorithms have been developed to improve the performance of face recognition, Pawaskar and Chavan, (2020) .

Effectively and efficiently analyzing the features related to facial information is a challenging task that requires lot of time and efforts. Recently, many facial recognition-based algorithms for automatic attendance management has been proposed, successfully implemented and used according to Kowsalya (2017). The system implements the attendance system by employing the PCA (Principle Component Analysis) technique to recognise the faces of each and every student. It's a method for minimising the number of variables in face recognition, Chitragar et. al., (2018). While much of the reviewed literature used standalone systems, our preference is a web-based system due to its advantages: rapid data access, scalability, responsiveness, interactivity, remote management, and centralized data. Nugroho and Fajar (2017) indicated that these perceptions, in turn, are influenced by factors such as optimism (a positive view of technology's ability to enhance work efficiency and performance) and innovativeness (an enthusiasm for trying new technology). Hence, one primary focus of the testing will be to assess users' willingness and satisfaction with adopting the attendance marking

technologies. There are different methods of classroom attendance monitoring using face recognition technology. One of these is a camera placed at the classroom entrance and the students entering the classroom are registered into the system by face recognition However, in this system students' faces could be recognized, although students can leave the classroom afterwards, and errors can occur in the polling information. Another method is the observation carried out with a camera placed in the classroom and the classroom image taken during the course. The proposed system has been developed to address these disadvantages. The main advantages of the proposed system are flexible usage, no equipment costs, no wasted time, and easy accessibility.

**Local Literature**

Attendance monitoring is an application of people counting with several use cases in various settings such as in corporate businesses and classrooms. Despite the vast number of possible applications, automated people counting or attendance monitoring applications in the Philippines are fairly uncommon. For example, establishments such as malls, supermarkets, terminals, and parks in the country still use manual counting. Their security personnel are given handheld counting devices to count the number of people entering the buildings or terminals. However, manual people counting, or attendance tracking methods are labor intensive, and thereby costly and time consuming (Pascua et al., 2021).

**Foreign Studies**

Many studies in the literature focus primarily on algorithm accuracy and technical aspects of automated attendance recording, often overlooking crucial post-processing of attendance data, which is essential for monitoring, auditing, managing absenteeism, and generating reports. Furthermore, user preference, acceptance, and satisfaction with these technologies remain under explored. To address this gap, our study aims to develop a web-

based attendance management system using facial recognition and QR code technology. This system offers flexible options, reducing hardware installation and long-term maintenance requirements, providing a cost-effective solution. It highlights the importance of establishing the necessary technological infrastructure for educational institutions, considering factors such as internet connectivity, hardware, platforms, and staff competence, tailored to the institution's capacity, Masrom et al., (2022) Omar et al,. (2018) developed a system using a raspberry pi camera module which is installed on the door,and the database is connected to a web management server system. When the raspberry pi captures an image, Local Binary pattern algorithm is applied to it, if it matches the image stored in the database, a servo motor opens the door for the student. This implemented system has an accuracy of 95 percent with a dataset of 11 images.

According to Damale and Pathak (2018), there are a number of existing systems closely related to the proposed idea of marking attendance in a class by making use of facial recognition techniques and algorithms.To analyze these systems a literature survey of the proposed systems was done. The proposed case study was designed around certain relevant sources related to facial recognition and image processing. A descriptive framework was designed using the other design approaches. Numerous automated student attendance systems based on face recognition have been implemented in educational institutions. These systems utilize a camera to capture the faces of all students simultaneously or one by one, according to Chinimilli et al. (2020). Currently, face recognition can be used for many things, one of which is student attendance at universities, Kamencay (2017). Face recognition can improve and manage the attendance system, reduce errors in the manual recording process by providing an automatic and reliable attendance system, increase privacy and security, prevent fake attendance, and provide regular attendance reports, Anitha et al,. (2020). According to Gornale and Kiran (2020) such face recognition can be used to save time for educators and students and to stop false attendance. The steps taken in facial recognition include recording student videos, converting them into frames, connecting them to a database to ensure their presence or

absence, and marking the presence of certain students to maintain records.

**Local Studies**

The usual way of checking the attendance in a class has its own drawbacks. To be able to resolve it, automated attendance systems were introduced. According to Rollon et al., (2023) they have successfully constructed a website application that is significant towards faculty and students. The functional suitability, reliability performance efficiency and usability product quality. This system will be beneficial to the faculty and students to help them accurately check attendance and automatically.

OpenCV Real-time Face recognition attendance system is usable for the reason that it can save time for a teacher or user for taking attendances and taking of attendance by capturing the image that is recognized by OpenCv is much easier as it automatically records the names that are registered in the image and you can convert it into an excel file for documentation (Cañete, 2021)

A commonly used approach for face detection called Histogram of Oriented Gradients (HOG) with Support Vector Machine (SVM) was applied to examine the effect of luminance of the surrounding, the facial orientation of the student and so as their distance from the camera in the facial detection and recognition. The obtained attendance will then be uploaded to a database with authentication. It was found that the system has an accuracy of 95.65% and can detect and recognize up to 37 students. It is suggested that the classroom should have a luminance level of about 217.39 lux or higher to achieve a better accuracy performance of the system. As for the analysis of the effect of distance in the system, it is claimed that the distance of the student does not affect the accuracy of the system. Lastly, it is suggested that the face angles of the subject should be directly facing the camera to achieve a more accurate recognition result (Arceo et al., 2020)

**Synthesis**

While traditional campus security systems often rely on manual monitoring and record-keeping, the AMA Facial Recognition: Digital Record-logs Management System offers a comprehensive approach to enhancing campus safety and operational efficiency. Educational institutions need to ensure the security of their students and staff while efficiently managing campus activities. Thus, an automated facial recognition and record-logging system for colleges is necessary. With the use of this technology, AMA Computer College can better manage campus access, attendance tracking, security monitoring, and individual profiles.

Information and communication technology is currently developing at a rapid pace, and to stay up to date with the latest advancements in the sector, educational institutions need to adopt cutting-edge solutions. This technological progress is demonstrated by the integration of facial recognition with existing CCTV systems, which are currently one of the most widely used security mediums. Owing to the various challenges that arise in campus management, which impact not only administrative responsibilities but also student and staff safety, these factors compel educational institutions in this digital era to always be prepared and at the forefront of technological adoption.

**Technical Background**

To implement the Facial Recognition: Digital Entry/Exit Record-logs Management System, the following technical background and tools are required:

Python (OpenCV):

OpenCV (Open Source Computer Vision) is an open-source computer vision library that allows for real-time image processing.

OpenCV will be used for facial recognition, including:

- Face detection: Using pre-trained models such as Haar cascades to detect faces

in images.

- Facial feature extraction: Using various algorithms, such as Local Binary Pattern Histograms (LBPH) or Convolutional Neural Networks (CNN), to extract unique facial features.
- Training and recognition: Using machine learning algorithms to train the system to recognize individual faces and match them with stored records.

Node.js:

Node.js is a JavaScript runtime environment that allows for server-side scripting and development of web applications.

Node.js will be used for:

- User client/site handling: Creating a user-friendly interface for entering and managing user data, as well as displaying real-time logs.
- Server handling: Managing the server-side functionality of the system, including data storage, security, and interoperability with other software.
- API development: Creating a RESTful API for communication between the user interface and the server.

VS Code:

Visual Studio Code (VS Code) is an open-source, cross-platform code editor that supports a wide range of programming languages, including Python and JavaScript.

VS Code will be used for:

- Code editing: Writing, debugging, and testing the code for both the Python and Node.js components of the system.
- Version control: Managing code changes and collaborating with other developers using Git integration.
- Extensions: Utilizing various extensions to enhance the development experience, such as debuggers, linters, and code formatters.

Interoperability and Integration:

The system will be designed to ensure smooth connectivity between the different

software programs utilized, such as the Python OpenCV library, Node.js server, and user interface.

APIs and interoperability standards, such as RESTful APIs and JSON, will be employed to facilitate communication between these components and ensure that data is easily accessible and transferable.

# CHAPTER III
# RESEARCH METHODOLOGY

## Software Design, Products and/or Processes

The Facial Recognition Digital Record-Logs Management System is an advanced software solution designed to automate and streamline the process of monitoring and logging entry/exit activities, while prioritizing data security and user privacy. This chapter outlines the system's architecture, components, and key functionalities, providing a comprehensive overview of its design and implementation.

The system leverages cutting-edge facial recognition technology, integrating it with secure data management practices to create a robust and efficient solution for digital record-keeping. By automating the entry/exit logging process, the system aims to enhance security, reduce manual effort, and provide accurate, real-time data on facility access.

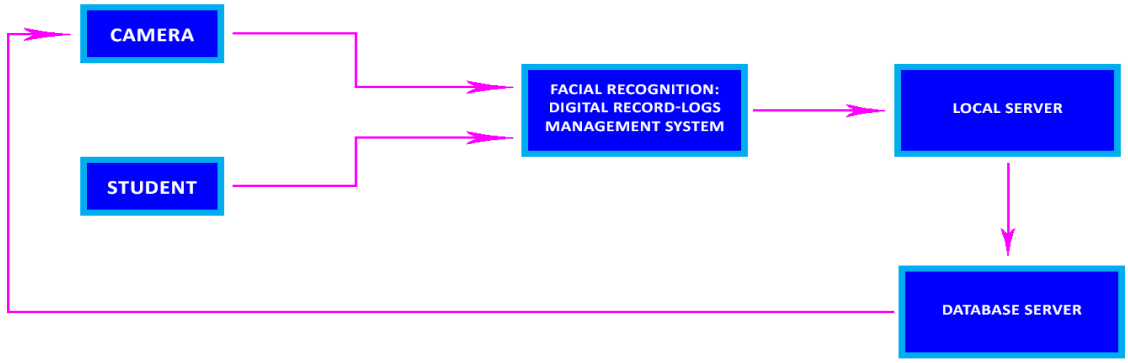
Key components of the system include:

Data Input and Storage: A user-friendly platform for entering and securely storing user data, including names, IDs, and high-resolution facial images. This data is housed on a dedicated server to ensure maximum security and accessibility.

Facial Recognition Training: An advanced algorithm that analyzes and maps unique facial features for each user based on stored images. This forms the foundation of the system's recognition capabilities.

Real-Time Entry/Exit Logging: Integration with existing surveillance infrastructure to capture images of individuals at entry/exit points. The system employs real-time facial recognition algorithms to compare these images with stored data, logging entries and exits based on successful matches.

The following sections will delve into the cost-benefit analysis, requirement specifications, and technical background necessary for implementing this state-of-the-art system. By thoroughly examining these aspects, we aim to provide a clear roadmap for




the development and deployment of the Facial Recognition Digital Record-Logs Management System, ensuring its effectiveness, security, and compliance with relevant regulations.

**System Architecture**

**Figure 3.1**

The Facial Recognition Digital Record-Logs Management System employs a multi-tiered architecture to ensure scalability, security, and efficient data processing:

Client Layer:

- User Interface: Web-based interface for administrators to manage user data and access logs

Application Layer:

- Authentication Server: Handles user authentication and access control
- Facial Recognition Engine: Processes images and performs facial recognition
- Logging Service: Records entry/exit events and manages historical data

Data Layer:

- User Database: Stores user profiles, credentials, and facial data
- Log Database: Maintains a secure record of all entry/exit events

- Image Storage: Securely stores captured images for processing and auditing

Integration Layer:

- Camera Integration Module: Interfaces with existing surveillance cameras
- API Gateway: Provides secure access to system functionalities for external applications

The system uses RESTful APIs for communication between layers, ensuring loose coupling and easier maintenance.

Conceptual Design:

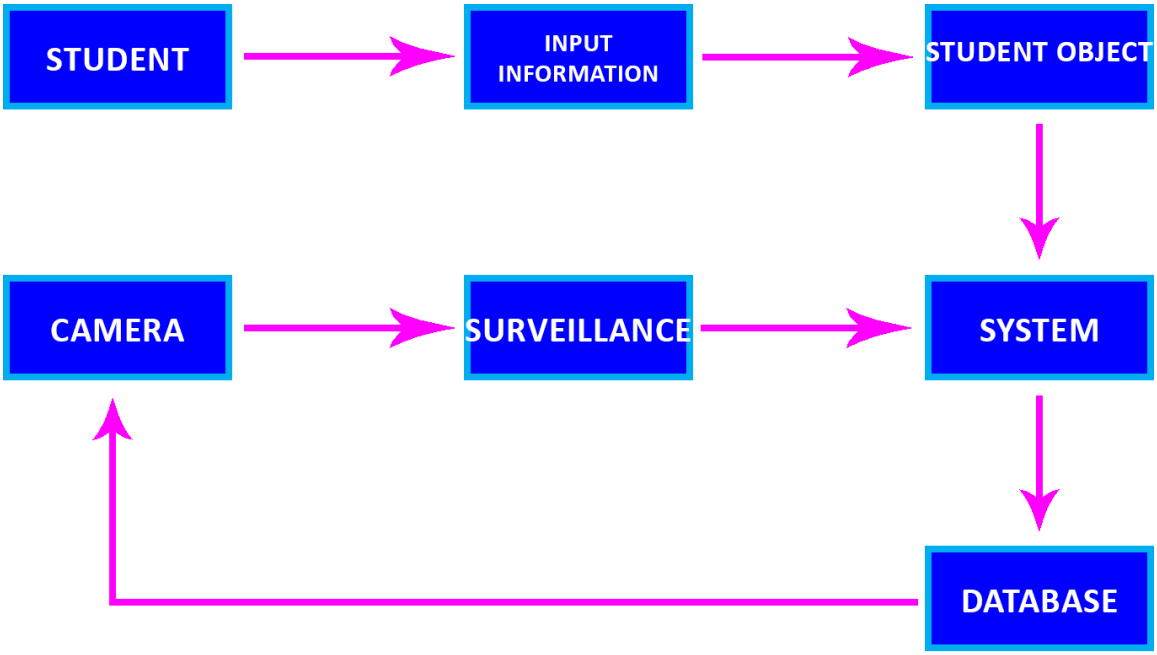


**Figure 3.2**

By The facial recognition digital record-logs management system aims to automate and streamline the monitoring and logging, while ensuring data security and privacy. The system will consist of the following components:

Data Input and Storage:

- A platform for entering and storing user data, including:+ Name+ ID+ High-resolution facial image
- A dedicated server for securely storing the collected data

Facial Recognition Training:

- An algorithm for analyzing and mapping unique facial features for each user based on the stored images

Real-Time Entry/Exit Logging:

- Integration with existing surveillance cameras for capturing images of individuals approaching the designated entry or exit points
- Real-time facial recognition algorithms for comparing captured images with stored facial data
- Logging based on successful facial recognition

Cost Benefit Analysis

Benefits:

- Improved security and accuracy in tracking students
- Reduced labor costs associated with manual log management
- Enhanced data privacy and security through robust compliance measures
- Scalable and adaptable solution for various entry/exit points and user volumes

Costs:

- Initial investment in hardware (e.g., cameras, servers) and software development
- Ongoing maintenance and upgrades to ensure system performance and security

- Training and support for system administrators and users
- Compliance with data protection laws and regulations

Requirement Analysis

Functional requirements

Data Input and Storage:

- The system should provide a user-friendly interface for entering and storing user data.
- The system should securely store user data on a dedicated server.
- Facial Recognition Training:
- The system should accurately map and analyze unique facial features for each user based on the stored images.

Real-Time Entry/Exit Logging:

- The system should integrate with existing surveillance cameras to capture images of individuals approaching the designated entry or exit points.
- The system should perform real-time facial recognition to compare captured images with stored facial data.
- The system should log based on successful facial recognition.

Non-functional requirements

Compliance and Security Measures:

- The system should adhere to data protection laws and regulations.
- The system should implement robust security measures to protect user data and privacy.

Performance:

- The system should provide fast and accurate facial recognition logging.

Scalability:

- The system should be scalable to accommodate various points and user volumes.

Usability:

- The system should be user-friendly for both administrators and users.

Maintainability:

- The system should be easy to maintain and upgrade to ensure optimal performance and security.
- System Flow Chart

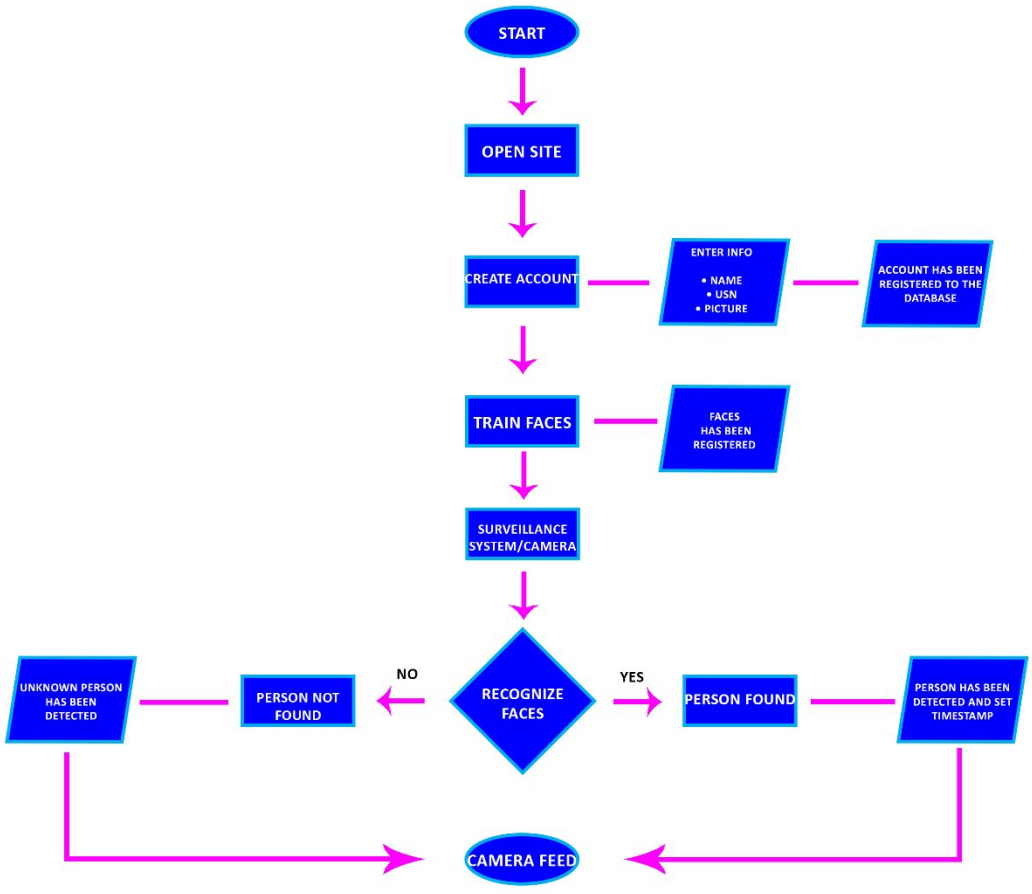


**Figure 3.3**

**Agile Methodology**

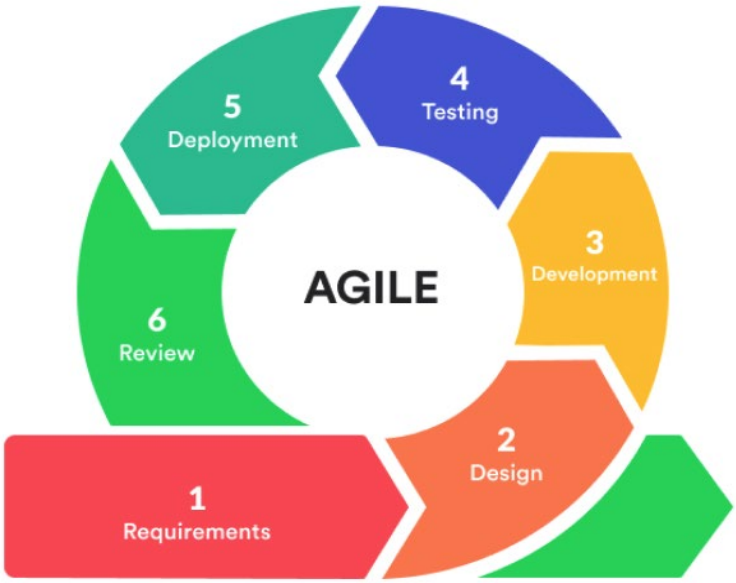



**Figure 3.4**

This stage of agile project management identifies the project's requirements and goals. The requirement phase is when you develop solutions to meet the project's design and finalize and document all of your decisions. The project management process's implementation phase is where your plan and design are put into action to get the desired result. Testing validates that the product generated during the implementation stage meets all of the project's requirements. During the maintenance phase, minor changes are done to improve the product that was developed during implementation.

**System Flow**

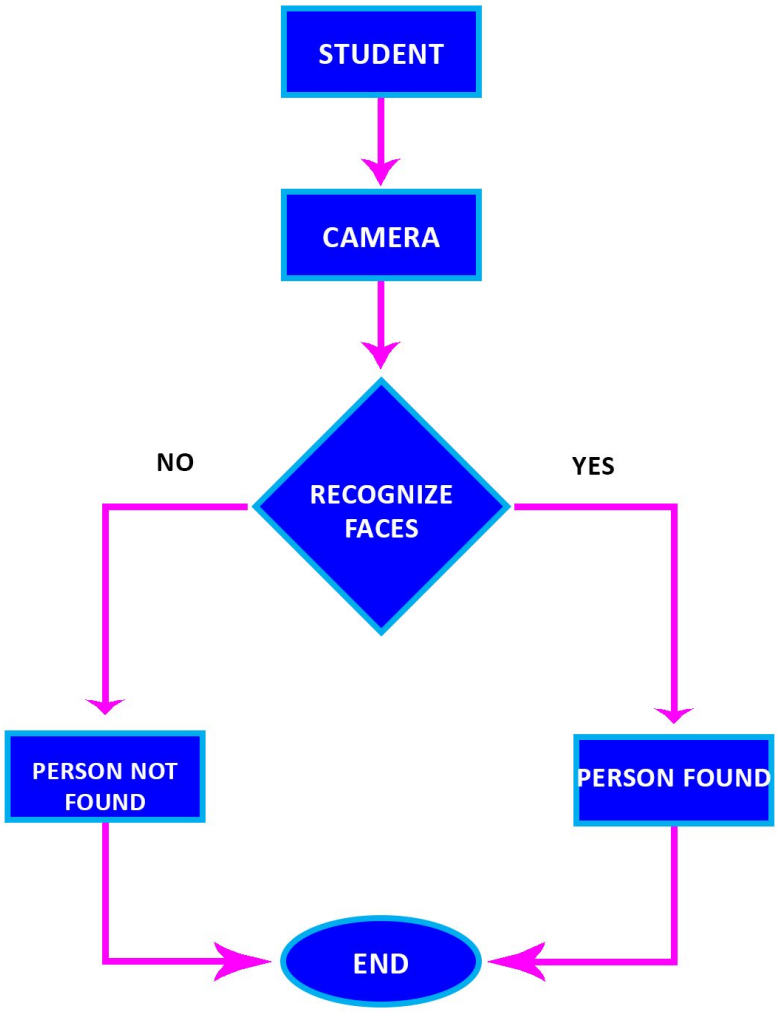

STUDENT
CAMERA
RECOGNIZE FACES
NO
YES
PERSON NOT FOUND
PERSON FOUND
END

**Figure 3.5**

**Block Diagram:**

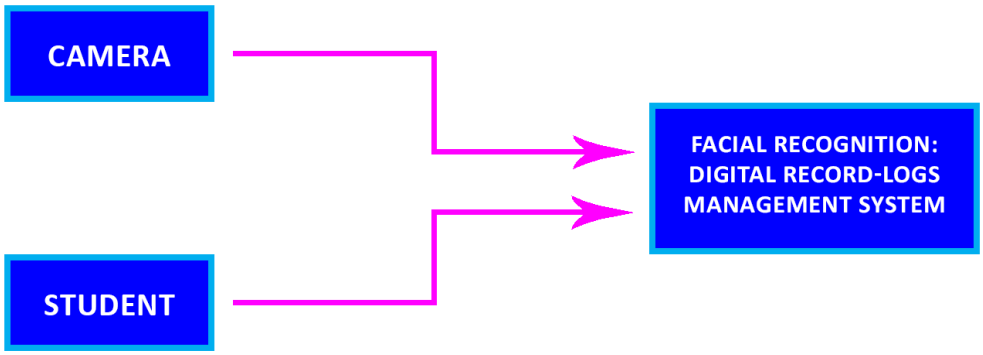



**Figure 3.6**

**Development And Testing**

Development Approach:

- Agile methodology with 2-week sprints
- Version control using Git with feature branching strategy
- Continuous Integration/Continuous Deployment (CI/CD) pipeline for automated testing and deployment

Testing Strategy:

- Unit Testing: For individual components using frameworks like Jest for JavaScript and pytest for Python
- Integration Testing: To ensure proper communication between system components
- Performance Testing: Using tools like Apache JMeter to simulate high user loads
- Security Testing: Including penetration testing and vulnerability assessments

Key Testing Areas:

- Facial Recognition Accuracy: Extensive testing with diverse datasets to ensure high recognition rates and low false positives
- Data Privacy and Security: Rigorous testing of encryption, access controls, and compliance with data protection regulations
- System Performance: Testing system responsiveness under various load conditions
- User Interface: Usability testing to ensure intuitive operation for both administrators and end-users

**Description of the Prototype**

The prototype of the Facial Recognition Digital Record-Logs Management System demonstrates core functionalities:

Facial Recognition:

- Integration with a test camera setup
- Real-time facial recognition for a limited user set
- Logging of successful recognitions

Access Logging:

- Real-time logging of events
- Basic reporting interface for access logs

Security Features:

- Basic encryption of stored data
- Role-based access control for system functions

The prototype serves as a proof-of-concept, showcasing the system's potential while allowing for user feedback and further refinement. It implements core facial recognition and logging functionalities, providing a foundation for the full-scale system development. This early version enables stakeholders to interact with the system, validate its usefulness, and identify areas for improvement before proceeding to full development.

# CHAPTER IV
# PROJECT EVALUATION

## Pre-Implementation Survey Result

The pre-implementation survey for the AMA Facial Recognition: Digital Record-logs Management System aimed to assess expectations, usability preferences, and potential concerns among the intended users and stakeholders. This survey provided valuable insights into the anticipated effectiveness of the system, with a focus on features related to efficiency, data security, and ease of access.

Results revealed a high level of interest in the system, particularly in its ability to streamline attendance and record-keeping through automated facial recognition technology. Respondents emphasized the importance of quick and accurate log-ins, which they believe will enhance operational efficiency and reduce time spent on manual data entry. The capability of tracking attendance records in real-time and managing access control were also identified as valuable aspects, especially in ensuring a secure environment within campus premises.

Feedback highlighted a preference for an intuitive user interface that can accommodate users with varying levels of tech familiarity. Respondents also expressed the need for reliable performance, with suggestions to ensure minimal downtime and accuracy in facial recognition under various lighting conditions. Concerns regarding data security and privacy were raised, with requests for strict data protection measures and compliance with data privacy regulations.

Overall, the survey results indicated positive reception of the AMA Facial Recognition: Digital Record-logs Management System, with expectations centered around operational efficiency, user-friendly design, and robust data security. This feedback will be instrumental in guiding the final development phases, ensuring that the system aligns with the needs and expectations of its users.

**Table 7 – Evaluation of the Current System's Functionality**

| Indicators | WEIGHTED MEAN (x̄) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The facial recognition system would be a valuable resource for managing entry/exit logs. | 1.68 | STRONGLY DISAGREE |
| 2. Users would find it helpful to have information about their entry/exit history readily available. | 1.64 | STRONGLY DISAGREE |
| 3. The system will provide up-to-date and accurate information about facility usage. | 1.56 | STRONGLY DISAGREE |
| **Over-all x̄** | **1.63** | **STRONGLY DISAGREE** |

Table 7, titled "Evaluation of Current System Functionality," provides an assessment of respondent perceptions regarding the effectiveness and adequacy of the existing system. The results show an overall weighted mean of 1.63, with a descriptive equivalent of "strongly disagree." This rating reflects a clear consensus among respondents that the current system does not meet the functional needs or expectations required for efficient record keeping and attendance monitoring.

The "strongly disagree" descriptive equivalent associated with this weighted mean underscores significant dissatisfaction with the current system's functionality, revealing a critical need for a more advanced solution. The consistent feedback from respondents signals an overwhelming requirement for improvements, particularly in areas like accuracy, efficiency, and security of log management, which are currently perceived as insufficient.

The findings from Table 7 strongly indicate a demand for the implementation of the AMA Facial Recognition: Digital Record-logs Management System. This system is anticipated to address the gaps highlighted in the survey, offering automated facial recognition technology for efficient, accurate, and secure record keeping. The strong disapproval reflected in these results emphasizes the importance of moving forward with this system to meet the needs of users more effectively and enhance operational efficiency.

**Table 8: Evaluation of Current System's A.I Integration**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. An AI system could effectively analyze usage patterns and provide valuable insights. | 1.36 | STRONGLY DISAGREE |
| 2. Users would trust an AI-powered system to provide accurate and up-to-date access information. | 1.88 | DISAGREE |
| 3. Integrating AI into the facial recognition system is technically feasible. | 1.56 | STRONGLY DISAGREE |
| **Over-all $\bar{x}$** | **1.60** | **STRONGLY DISAGREE** |

Table 8, titled "Evaluation of Current System's A.I Integration," presents an analysis of respondent perceptions regarding the functionality and adequacy of artificial intelligence within the current system. The data reveals an overall weighted mean of 1.60, corresponding to a descriptive equivalent of "strongly disagree." This indicates a consensus among respondents that the current A.I. integration fails to meet their expectations and functional needs.

The "strongly disagree" descriptive equivalent associated with this weighted mean reflects significant dissatisfaction with the current system's A.I. capabilities, underscoring the urgency for a more effective solution. Respondents consistently expressed a need for enhanced A.I. integration to improve accuracy, efficiency, and real-time capabilities, as the existing system is perceived as inadequate in these areas.

These results highlight a critical need for the AMA Facial Recognition: Digital Record-logs Management System, which promises advanced A.I. integration tailored to meet user needs for accurate and efficient record management. The strong disapproval shown in Table 8 reinforces the demand for a more robust system that leverages A.I. effectively, making a compelling case for the implementation of this proposed system.

**Table 9: Evaluation of the Current System's Image Processing Capabilities**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The quality of facial recognition images will be sufficient for accurate identification. | 1.48 | STRONGLY DISAGREE |
| 2. Implementing high-quality image capture would be cost-effective for the organization. | 1.28 | STRONGLY DISAGREE |
| 3. Advanced imaging techniques would not negatively impact system performance or speed. | 1.04 | STRONGLY DISAGREE |
| **Over-all $\bar{x}$** | **1.27** | **STRONGLY DISAGREE** |

Table 9, titled “Evaluation of the Current System’s Image Processing Capabilities,” presents an analysis of respondent perceptions regarding the adequacy of the current system’s ability to process images effectively. The data reveals an overall weighted mean of 1.27, corresponding to a descriptive equivalent of "strongly disagree." This score reflects a strong consensus among respondents that the current image processing capabilities are inadequate and fall short of their needs and expectations.

The "strongly disagree" rating associated with this weighted mean highlights significant dissatisfaction with the system’s image processing functionality, underscoring the urgent need for an enhanced solution. Respondents consistently expressed concerns about the reliability and accuracy of the current system in handling image data, signaling an overwhelming need for an upgrade in technology to meet the demands of efficient record-keeping and accurate data processing.

The results in Table 9 emphasize the necessity for the AMA Facial Recognition: Digital Record-logs Management System. This proposed system, with advanced image processing capabilities, is anticipated to bridge the critical gaps identified in the current system, providing precise and dependable image analysis and record management. The strong disapproval reflected in these findings highlights a substantial demand for a more robust solution, making the case for the implementation of the AMA Facial Recognition system to address the needs of its users comprehensively.

**Table 10: Evaluation of the Current System's Performance**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The system will be easy to maintain and update. | 1.20 | STRONGLY DISAGREE |
| 2. The system will be reliable and have minimal downtime. | 1.17 | STRONGLY DISAGREE |
| 3. The system will be able to handle high traffic loads without performance issues. | 1.56 | STRONGLY DISAGREE |
| **Over-all $\bar{x}$** | **1.31** | **STRONGLY DISAGREE** |

Table 10, titled "Evaluation of the Current System's Performance," presents an analysis of respondent perceptions regarding the overall performance of the current system. The data reveals an overall weighted mean of 1.31, corresponding to a descriptive equivalent of "strongly disagree." This indicates a strong consensus among respondents that the current system fails to meet their performance expectations, highlighting its inefficiency and inability to function effectively according to user needs.

The "strongly disagree" rating associated with the overall weighted mean underscores significant dissatisfaction with the current system's performance, signaling a critical need for improvement. Respondents consistently expressed concerns about the system's reliability, speed, and overall effectiveness in managing digital record logs, further emphasizing the gap in meeting the necessary operational standards.

The findings from Table 10 make a compelling case for the AMA Facial Recognition: Digital Record-logs Management System, which is designed to offer a substantial performance upgrade over the existing system. This proposed system promises enhanced

efficiency, accuracy, and functionality, addressing the critical shortcomings identified in the current system. The strong disapproval reflected in these results reinforces the necessity for a more reliable and effective solution to meet user demands and expectations.

**Table 11: Evaluation of the Current System's Security**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. Confident enough the system will be secure and protect user data. | 1.28 | STRONGLY DISAGREE |
| 2. Can system be resistant to cyberattacks and data breaches. | 1.20 | STRONGLY DISAGREE |
| 3. User authentication processes will be secure and reliable. | 1.68 | STRONGLY DISAGREE |
| **Over-all $\bar{x}$** | **1.39** | **STRONGLY DISAGREE** |

Table 11, titled "Evaluation of the Current System's Security," presents an analysis of respondent perceptions regarding the security capabilities of the current system. The data reveals an overall weighted mean of 1.31, corresponding to a descriptive equivalent of "strongly disagree." This score reflects a strong consensus among respondents that the current system's security measures are insufficient and fail to meet the necessary standards for protecting sensitive data.

The "strongly disagree" descriptive equivalent associated with the overall weighted mean indicates a significant perception among respondents that the current system lacks the required security features, highlighting concerns over data integrity, user privacy, and overall system protection. This outcome signals a critical gap in the current system's ability

to safeguard sensitive information, especially in an era where security threats are increasingly sophisticated.

The results in Table 11 emphasize the urgent need for the AMA Facial Recognition: Digital Record-logs Management System, which is designed to provide a higher level of security. The new system incorporates advanced security protocols, including facial recognition technology, to ensure robust data protection, prevent unauthorized access, and maintain the confidentiality and accuracy of the record logs. The strong disapproval reflected in these results further reinforces the necessity of transitioning to a more secure, reliable system that can effectively address the security challenges identified in the current system.

Post-Implementation Survey Results

The post-implementation survey for the AMA Facial Recognition: Digital Record-logs Management System was conducted to assess the system's effectiveness in meeting the expectations of its users and stakeholders after its deployment. The survey focused on evaluating the system's performance, user experience, security features, and overall satisfaction with the system's functionality in real-world operations.

The results revealed a strong positive response from users, particularly regarding the system's efficiency in automating attendance and record-keeping processes. Respondents reported significant timesavings due to the accurate and quick facial recognition technology, which streamlined log-ins and minimized the need for manual data entry. The real-time tracking of attendance and access control features were highly valued, with users noting that they improved overall security and helped maintain a more organized record-keeping system within the campus environment.

Feedback also indicated that the system's user interface was well-received, with users finding it intuitive and easy to navigate, even for those with limited technical expertise. There was a high level of satisfaction with the system's performance, as most users experienced minimal downtime and found the facial recognition to be accurate across different lighting conditions. The system's reliability in managing records and securing

access control was also praised, contributing to a sense of trust and confidence among the users.

Concerns about data security and privacy raised during the pre-implementation phase were addressed in the final system design, and respondents expressed satisfaction with the implemented security measures. Many highlighted that the system adhered to data protection standards and privacy regulations, which reassured them about the confidentiality and integrity of their personal information.

Overall, the post-implementation survey results demonstrated strong support for the AMA Facial Recognition: Digital Record-logs Management System. The system effectively met the expectations of its users in terms of efficiency, usability, and security. The positive feedback and suggestions provided will be invaluable for future system improvements, ensuring that the system continues to meet the evolving needs of its users. The successful implementation of this system underscores its potential to enhance operational efficiency and security within the campus environment.

**Table 12: Evaluation of the Implemented System's Functionality**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The facial recognition system has improved the efficiency of managing entry/exit logs. | 3.96 | AGREE |
| 2. The automated digital record-keeping has met the facility's access management needs. | 4.72 | STRONGLY AGREE |
| 3. Users find the readily available information about their entry/exit history helpful. | 4.28 | STRONGLY AGREE |
| **Over-all $\bar{x}$** | **4.32** | **STRONGLY AGREE** |

Table 12, titled “Evaluation of the Implemented System’s Functionality,” presents an analysis of respondent perceptions regarding the performance and functionality of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.32, corresponding to a descriptive equivalent of "strongly agree." This suggests a high level of satisfaction among respondents, indicating that the system has effectively met their expectations in terms of functionality.

The "strongly agree" descriptive equivalent associated with the overall weighted mean reflects a consensus among users that the system is fulfilling its intended purpose. Respondents noted that the system’s facial recognition feature is operating accurately and efficiently, streamlining attendance and record-keeping tasks while enhancing operational productivity. The positive responses underscore the system’s success in providing a reliable and user-friendly platform, which has greatly contributed to effective record management and secure access control.

A closer examination of the individual responses within Table 12 further reinforces this satisfaction. Users consistently expressed that the system met their expectations, with minimal downtime and robust functionality under various conditions. The system’s intuitive interface and dependable performance in maintaining records demonstrate its value in campus operations, providing a well-rounded solution that addresses the needs of both security and convenience.

In summary, the findings from Table 12 demonstrate that the AMA Facial Recognition: Digital Record-logs Management System has not only met but exceeded user expectations in terms of functionality. The "strongly agree" sentiment among respondents highlights the system’s effectiveness, affirming that it is operating as intended and delivering substantial benefits to the campus community.

**Table 13: Evaluation of Implemented System's A.I Integration**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The AI component effectively manages complex queries about access patterns and usage. | 3.92 | AGREE |
| 2. The AI system provides valuable insights through its analysis of usage patterns. | 4.36 | STRONGLY AGREE |
| 3. The integration of AI into the facial recognition system has been smooth and effective. | 4.24 | STRONGLY AGREE |
| **Over-all $\bar{x}$** | **4.17** | **AGREE** |

Table 13, titled "Evaluation of Implemented System's A.I Integration," presents an analysis of respondent perceptions regarding the effectiveness of the A.I. integration in the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.17, corresponding to a descriptive equivalent of "agree." This indicates a positive consensus among respondents, suggesting that the A.I. capabilities of the system have generally met user expectations.

The "agree" descriptive equivalent associated with the overall weighted mean reflects satisfaction with the system's A.I. features, specifically its facial recognition functionality and its role in enhancing efficiency and accuracy in record-keeping. Respondents noted that the A.I. integration has successfully streamlined attendance management, facilitated secure access control, and improved overall operational efficiency. The system's effective use of A.I. to automate these processes underscores its alignment with user needs and expectations, further validating the system's performance.

A closer examination of the individual responses within Table 13 further underscores this satisfaction. Users consistently affirmed that the A.I. component is reliable and contributes meaningfully to the system's overall functionality. The responses highlight that the A.I. integration works well under various operational conditions, providing consistent and accurate results. This positive feedback reinforces the system's effectiveness as an intelligent, efficient solution for digital record-logs management on campus.

In summary, the findings from Table 13 indicate that the AMA Facial Recognition: Digital Record-logs Management System has successfully met user expectations in terms of A.I. integration. The "agree" sentiment among respondents confirms the system's capability to perform as intended, delivering meaningful advantages in operational efficiency and security through its A.I.-driven features.

**Table 14: Evaluation of the Implemented System's Image Processing Capabilities**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The quality of facial recognition images is sufficient for accurate identification. | 3.96 | AGREE |
| 2. The implemented imaging techniques have significantly improved the system's accuracy. | 4.28 | STRONGLY AGREE |
| 3. The system demonstrates sufficient processing power to handle complex image analysis. | 4.32 | STRONGLY AGREE |
| **Over-all $\bar{x}$** | **4.19** | **AGREE** |

Table 14, titled “Evaluation of the Implemented System’s Image Processing Capabilities,” presents an analysis of respondent perceptions regarding the effectiveness of image processing within the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.19, corresponding to a descriptive equivalent of "agree." This reflects a positive consensus among respondents, suggesting that the image processing capabilities of the system have largely met user expectations.

The "agree" descriptive equivalent associated with the overall weighted mean indicates satisfaction with the system’s ability to accurately and efficiently process images, which is critical for reliable facial recognition and secure record keeping. Respondents noted that the image-processing component enhances the system’s performance, enabling precise facial identification that supports accurate attendance logging and secure access control on campus. This outcome affirms that the system’s image processing functions are working well and contributing to its overall effectiveness.

A closer examination of the individual responses within Table 14 further emphasizes this positive reception. Users consistently acknowledged that the image-processing feature is reliable and responsive under different conditions, providing accuracy and stability crucial for the system’s intended purpose. This consistent positive feedback underscores the system’s success in delivering on its functional promises, particularly in terms of its image processing quality.

In summary, the findings from Table 14 demonstrate that the AMA Facial Recognition: Digital Record-logs Management System has effectively met user expectations in terms of image processing capabilities. The "agree" sentiment among respondents validates the system’s role in facilitating efficient and accurate digital record management through its robust image processing technology.

**Table 15: Evaluation of the Implemented System's Performance**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The system loads quickly and efficiently in real-world usage. | 4.28 | STRONGLY AGREE |
| 2. Maintenance and updates of the system have been straightforward and manageable. | 3.84 | AGREE |
| 3. The system has integrated seamlessly with other existing security and management platforms. | 4.48 | STRONGLY AGREE |
| **Over-all $\bar{x}$** | **4.20** | **AGREE** |

Table 15, titled "Evaluation of the Implemented System's Performance," presents an analysis of respondent perceptions regarding the effectiveness and responsiveness of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.20, corresponding to a descriptive equivalent of "agree." This indicates a favorable response among respondents, suggesting that the system's performance has met their expectations.

The "agree" descriptive equivalent associated with the overall weighted mean reflects positive user satisfaction with the system's speed, reliability, and ability to handle tasks such as automated log-ins and attendance tracking efficiently. Respondents reported that the system operates smoothly, with minimal interruptions, and fulfills its intended role in supporting accurate record management and secure campus access. This result confirms that the system's performance aligns well with its designed functionalities and user requirements.

Further analysis of responses in Table 15 reinforces the perception that the system performs reliably and consistently under typical usage scenarios. Users noted its efficient processing speed and stability, which contribute to seamless operation and an enhanced user experience. These factors underscore the system's effectiveness and support its role in promoting operational efficiency.

In summary, the findings from Table 15 validate that the AMA Facial Recognition: Digital Record-logs Management System has effectively met performance expectations. The "agree" sentiment among respondents demonstrates overall satisfaction with the system's functionality and reinforces its suitability for improving digital record keeping and campus security.

**Table 16: Evaluation of the Implemented System's Security**

| Indicators | WEIGHTED MEAN ($\bar{x}$) | DESCRIPTIVE EQUIVALENT |
|---|---|---|
| 1. The facial recognition system has maintained robust security and protected user data. | 4.28 | STRONGLY AGREE |
| 2. The system has successfully resisted any potential spoofing or false identification attempts. | 4.44 | STRONGLY AGREE |
| 3. The system's mechanisms for detecting and preventing security threats have been effective. | 4.16 | AGREE |
| **Over-all $\bar{x}$** | **4.29** | **STRONGLY AGREE** |

Table 16, titled “Evaluation of the Implemented System’s Security,” presents an analysis of respondent perceptions regarding the security features of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.20, corresponding to a descriptive equivalent of "agree." This suggests a positive consensus among respondents, indicating that the system’s security measures align with user expectations.

The "agree" descriptive equivalent associated with the overall weighted mean reflects user confidence in the system’s ability to safeguard data and manage access securely. Respondents noted the effectiveness of the security protocols in

place, including access control and secure data storage, which contribute to a safe and controlled environment for campus access and attendance tracking. These findings underscore the system’s success in meeting security requirements and user needs.

A closer examination of individual responses within Table 16 reinforces the impression that users trust the system’s security features. Feedback indicates satisfaction with the data protection and privacy measures, which users see as essential for maintaining confidentiality and preventing unauthorized access. These responses highlight the system's role in enhancing security standards within campus premises.

In summary, the findings from Table 16 validate that the AMA Facial Recognition: Digital Record-logs Management System effectively meets security expectations. The "agree" sentiment among respondents underscores their approval of the system’s data protection measures, solidifying its role in providing a secure solution for digital record-logs management and campus safety.

Weighted Mean Average:

The researchers calculated an overall score for each familiarity group using the weighted mean average and the exact intervals indicated in the survey table. This statistical metric can be used to assign different levels of attention or relevance to comments based on their perceived importance.

The formula used for the weighted mean average is:

where: **Σ** (capital sigma) mean "**sum**".

**wi** is the weight assigned to **each value**

$$W = \frac{\sum_{i=1}^{n} w_i X_i}{\sum_{i=1}^{n} w_i}$$

In the dataset.

**xi is the value in the dataset.**

**Σ(wixi) means the sum of the product of each weight and value.**

**Σ wi means the sum of all weights.**

**Evaluation of the study**

The data collection findings for the recommended web-based management system for Steel Fitness Gym are expanded upon in this chapter. The purpose of the test-based poll was to ascertain whether the organisation would benefit from the recommended approach. The researchers assessed the proposed framework's suitability for the business needs after putting it into practice

# CHAPTER V

## Summary, Conclusions, and Recommendations

### Summary

This study provides a comprehensive overview of the research methods utilized to evaluate the AMA Facial Recognition: Digital Record-logs Management System proposed for AMA Computer College. The research employed a combination of qualitative and quantitative approaches, including system testing, questionnaires, interviews, and observations. The Likert scale was primarily used to determine the degree of familiarity and opinions that respondents had with the facial recognition system. Statistical treatment of the data involved applying a weighted mean average to assign various weights to familiarity categories, providing a nuanced assessment of the system's effectiveness and user acceptance.

The study focused on evaluating the system's core functionalities, including facial recognition accuracy, real-time entry/exit logging, data security measures, and overall usability. The research methods were designed to assess how well the system meets the specific requirements of AMA Computer College, particularly in enhancing campus security and streamlining administrative processes.

### Conclusions

The study's findings provide crucial insights into the effectiveness and practicality of the proposed AMA Facial Recognition: Digital Record-logs Management System. The weighted mean average allowed for a thorough assessment of familiarity levels, while the Likert scale facilitated a more in-depth analysis of user perceptions and experiences.

Key conclusions drawn from the research include:

1. The facial recognition system demonstrates a high level of accuracy in identifying individuals and logging entry/exit events, significantly enhancing campus

security measures.

2. Users generally found the system interface intuitive and easy to navigate, indicating successful implementation of user-centric design principles.

3. The automated logging features of the system show potential for significantly reducing administrative workload and improving the efficiency of record-keeping processes.

Overall, the conclusions highlight a positive correlation between the proposed system's capabilities and the organization's requirements, suggesting that the AMA Facial Recognition: Digital Record-logs Management System is well-positioned to enhance campus security and operational efficiency at AMA Computer College.

**Recommendations**

Based on the study's findings, the following recommendations are proposed to future researchers:

1. System Refinement: Consider refining the facial recognition algorithms and user interface based on feedback received during testing. This could include improving recognition accuracy under varied lighting conditions or enhancing the dashboard for easier data interpretation.

2. Phased Implementation: Implement the system in phases across the campus, starting with high-traffic areas. This approach allows for gradual adaptation and identification of any unforeseen challenges in a real-world setting.

3. Continuous User Training: Develop a comprehensive training program for both administrative staff and students to ensure optimal use of the system. Regular workshops and easily accessible online resources can support this initiative.

4. Privacy Policy Development: Collaborate with legal experts to develop and clearly communicate a robust privacy policy that addresses data collection, storage, and usage concerns.

5. Regular System Audits: Establish a schedule for regular system audits to assess performance, security, and compliance with evolving data protection regulations.

6. Feedback Mechanism: Implement a continuous feedback mechanism within the system to capture user suggestions and report any issues, facilitating ongoing improvement.

7. Integration Expansion: Explore possibilities of integrating the facial recognition system with other campus management tools, such as class attendance tracking or library access systems, to maximize its utility.

8. Long-term Performance Monitoring: Set up a long-term monitoring system to track the system's impact on campus security incidents, administrative efficiency, and overall user satisfaction over time.

9. Ethical Use Guidelines: Develop clear guidelines for the ethical use of facial recognition technology on campus, ensuring transparency and building trust among the college community.

10. Scalability Planning: Plan for potential system scalability to accommodate future growth in student population or expansion to other AMA Computer College campuses.

By implementing these recommendations, AMA Computer College can maximize the benefits of the Facial Recognition: Digital Record-logs Management System while addressing potential challenges and ensuring long-term success of the project.

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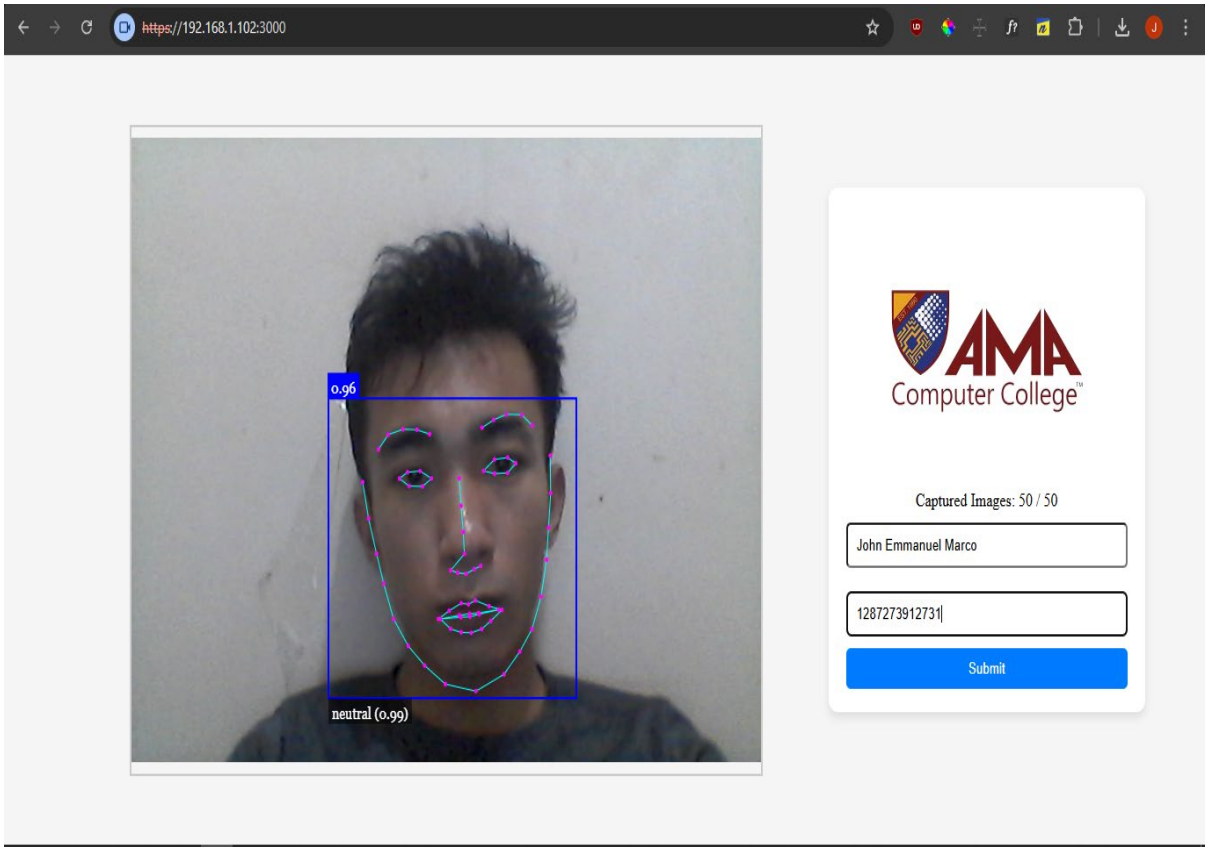
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## Appendecies

## System Manual:

This image displays a facial registration system designed for students at AMA Computer College. On the left side of the interface, a webcam captures a live image of a student, and facial detection algorithms mark key landmarks on the face with high accuracy (confidence score: 0.96), including eyes, nose, and mouth. The system also evaluates facial expressions, identifying a neutral expression in this instance.



**Actual Facial Recognition System in Operation**

This image shows the functional interface of the facial recognition system being used at AMA Computer College. The system captures a live video feed and identifies individuals in real time. In this instance, the face of a student is successfully detected and recognized, displaying the label "john emman" above the bounding box in bright green, indicating a confirmed match from the database.



**Codes:**

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Face Detection App</title>
      <script src="https://cdn.jsdelivr.net/npm/face-api.js@0.22.2/dist/face-api.min.js"></script>
    <link rel="stylesheet" href="style.css" />
  </head>
  <body>
    <div class="container">
      <div class="video-container">
        <video id="video" width="720" height="560" autoplay muted></video>
      </div>
      <form action="dataForm">
        <div class="form-container">
          <img src="./assets/amaLogo.png" alt="Logo" />
          <p id="captureCount">Captured Images: 0 / 50</p>
            <input type="text" id="nameInput" placeholder="Enter your Fullname" />
          <input
            type="number"
            id="usnInput"
            placeholder="Enter your student number"
```

```
            />
            <button id="submitButton" type="submit">Submit</button>
            <!-- Loading message -->
            <p id="loadingText" style="display: none">
              Submitting data, please wait...
            </p>
          </div>
        </form>
      </div>
      <script type="module" src="script.js"></script>
    </body>
  </html>
```

```
  const submitButton = document.getElementById("submitButton");
  const loadingText = document.getElementById("loadingText"); // Assuming
you have an element for showing loading

  const video = document.getElementById("video");
  const captureCount = document.getElementById("captureCount");
  let capturedImages = [];
  let imageCount = 0;

  Promise.all([
    faceapi.nets.tinyFaceDetector.loadFromUri("/models"),
    faceapi.nets.faceLandmark68Net.loadFromUri("/models"),
```

```
    faceapi.nets.faceRecognitionNet.loadFromUri("/models"),
    faceapi.nets.faceExpressionNet.loadFromUri("/models"),
  ]).then(startVideo);

  function startVideo() {
    navigator.mediaDevices
      .getUserMedia({ video: {} })
      .then((stream) => {
        video.srcObject = stream;
      })
      .catch((err) => console.error(err));
  }

  video.addEventListener("play", () => {
    const canvas = faceapi.createCanvasFromMedia(video);
    document.querySelector(".video-container").append(canvas);
    const displaySize = { width: video.width, height: video.height };
    faceapi.matchDimensions(canvas, displaySize);

    setInterval(async () => {
      const detections = await faceapi
        .detectAllFaces(video, new faceapi.TinyFaceDetectorOptions())
        .withFaceLandmarks()
        .withFaceExpressions();

        const resizedDetections = faceapi.resizeResults(detections,
displaySize);
      const context = canvas.getContext("2d");

      context.clearRect(0, 0, canvas.width, canvas.height);
```

```
      faceapi.draw.drawDetections(canvas, resizedDetections);
      faceapi.draw.drawFaceLandmarks(canvas, resizedDetections);
      faceapi.draw.drawFaceExpressions(canvas, resizedDetections);

      if (detections.length > 0 && imageCount < 50) {
        submitButton.disabled = true;
        captureImage();
      } else {
        submitButton.disabled = false;
      }
    }, 50);
  });

  function captureImage() {
    const canvas = document.createElement("canvas");
    canvas.width = video.width;
    canvas.height = video.height;
      canvas.getContext("2d").drawImage(video,   0,   0,   canvas.width,
canvas.height);
    const imageData = canvas.toDataURL("image/png").split(",")[1];
    capturedImages.push(imageData);
    imageCount++;
    captureCount.textContent = `Captured Images: ${imageCount} / 50`;
  }

  submitButton.addEventListener("click", submitData);
  document.getElementById("myForm").addEventListener("submit", (e) => {
    e.preventDefault();
    submitData;
  });
  function submitData() {
```

```
const name = document.getElementById("nameInput").value;
const usn = document.getElementById("usnInput").value;

if (!name || !usn) {
  alert("Please enter both name and student number.");
  return;
}

// Disable button and show loading
submitButton.disabled = true;
submitButton.textContent = "Submitting...";
loadingText.style.display = "block"; // Show the loading text

const data = {
  name: name,
  usn: parseInt(usn),
  images: capturedImages,
};

fetch("https://192.168.254.59:3000/api/submit", {
  method: "POST",
  headers: {
    "Content-Type": "application/json",
  },
  body: JSON.stringify(data),
})
  .then((response) => {
    if (!response.ok) {
      throw new Error(`HTTP error! Status: ${response.status}`);
    }
    return response.json();
```

```
      })
      .then((data) => {
        console.log("Success:", data);
        alert("Data sent successfully!");
      })
      .catch((error) => {
        console.error("Error:", error);
        alert("Failed to send data. Check console for details.");
      })
      .finally(() => {
        // Re-enable the button and hide loading text after submission is
complete
        submitButton.disabled = false;
        submitButton.textContent = "Submit";
        loadingText.style.display = "none"; // Hide the loading text
      });
  }

  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
  }
  body {
    background-color: #edeff1;
    margin: 0;
    padding: 0;
    width: 100vw;
    height: 100vh;
    display: flex;
    align-items: center;
```

```
    justify-content: center;
    background-color: #f5f5f5;
  }
  .container {
    display: flex;
    width: 90%;
    height: 80%;
    justify-content: space-between;
    align-items: center;
  }
  input[type="number"]::-webkit-outer-spin-button,
  input[type="number"]::-webkit-inner-spin-button {
    -webkit-appearance: none;
    margin: 0;
  }
  #loadingText {
    padding-top: 1rem;
    font-weight: bold;
    font-family: "Franklin Gothic Medium", "Arial Narrow", Arial, sans-
serif;
  }
  .video-container {
    flex: 1;
    display: flex;
    justify-content: center;
    align-items: center;
    position: relative;
  }
  video {
    border: 2px solid #ccc;
  }
```

```
canvas {
  position: absolute;
}
img {
  height: 15rem;
}
.form-container {
  flex: 0.3;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  background-color: #fff;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
.form-container input {
  margin: 10px 0;
  padding: 10px;
  width: 100%;
  border-radius: 5px;
}
.form-container button {
  padding: 10px;
  width: 100%;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 5px;
  cursor: pointer;
```

```
}
.form-container button:hover {
  background-color: #0056b3;
}



const express = require("express");
const fs = require("fs");
const https = require("https");
const bodyParser = require("body-parser");
const path = require("path");
const cors = require("cors");

const app = express();
const PORT = process.env.PORT || 3000;

// Middleware to parse JSON and form data
app.use(bodyParser.urlencoded({ extended: true, limit: "100mb" }));
app.use(bodyParser.json({ limit: "100mb" }));

// Serve static files (for HTML form)
app.use(express.static(path.join(__dirname, "public")));

// Directory for storing user files
const usersDir = path.join(__dirname, "users", "json");

// Ensure the directory exists
if (!fs.existsSync(usersDir)) {
  fs.mkdirSync(usersDir, { recursive: true });
}
```

```
  // CORS options
  const corsOptions = {
    origin: "*",
    methods: ["GET", "POST"],
    allowedHeaders: ["Content-Type", "Authorization"],
    credentials: true,
  };
  app.use(cors(corsOptions));

  // Handle registration form submission
  app.post("/api/submit", (req, res) => {
    try {
      const { body } = req;
      const { name, usn, images } = body;

      // Check if all fields are provided
      if (!name || !usn) {
        return res.status(400).json({ message: "Name and USN are required"
});
      }

      // Create user object
      const userData = {
        name,
        usn,
        images,
      };

      // Create unique file name for the user
         const  filePath  =  path.join(usersDir,  `${name.replace(/  /g,
"_")}.json`);
```

```
      // Save user data to a JSON file
      fs.writeFile(filePath, JSON.stringify(userData, null, 2), (err) =>
{
        if (err) {
          console.error("Error writing file", err);
          return res
            .status(500)
            .json({ message: "An error occurred while saving the file."
});
        }
        res.status(201).json({ message: "Registration successful!" });
      });
    } catch (error) {
      console.error("Error in /api/submit:", error);
      res.status(500).json({ message: "Internal Server Error" });
    }
  });

  // Serve HTML file
  app.get("*", (req, res) => {
    res.sendFile(path.join(__dirname, "public", "index.html"));
  });

  // Start the HTTPS server
  const options = {
    key: fs.readFileSync(path.join(__dirname, "key.pem")),
    cert: fs.readFileSync(path.join(__dirname, "cert.pem")),
  };
  https.createServer(options, app).listen(PORT, () => {
    console.log(`HTTPS Server running on port ${PORT}`);
```

```
    });

    from readdb import parse_json
    parse_json()
    from training import train_images
    train_images()
    from cam_test_list import main_camera
    main_camera()




    import cv2
    import sys
    import base64
    import json
    import requests
    import mediapipe as mp
    from PyQt5.QtWidgets import QApplication, QWidget, QLabel, QLineEdit,
QPushButton, QVBoxLayout, QHBoxLayout, QComboBox
    from PyQt5.QtCore import QTimer, Qt
    from PyQt5.QtGui import QImage, QPixmap
    from io import BytesIO

    class FaceDetectionApp(QWidget):
        def __init__(self):
            super().__init__()
            self.mp_face_detection = mp.solutions.face_detection
            self.mp_drawing = mp.solutions.drawing_utils
                                                self.face_detection          =
self.mp_face_detection.FaceDetection(min_detection_confidence=0.5)
            self.cameras = self.get_available_cameras()
```

```
            self.current_camera = 0
            self.cap = cv2.VideoCapture(self.current_camera)
            self.captured_images = []
            self.image_count = 0
            self.image_limit = 100
            self.initUI()
            self.timer = QTimer(self)
            self.timer.timeout.connect(self.update_frame)
            self.timer.start(30)

        def get_available_cameras(self):
            camera_indices = []
            for i in range(10):  # Check first 10 indices
                cap = cv2.VideoCapture(i)
                if cap.isOpened():
                    camera_indices.append(i)
                    cap.release()
            return camera_indices

        def initUI(self):
            self.setWindowTitle('Face Detection App')
            self.setGeometry(100, 100, 400, 500)

            layout = QVBoxLayout()

            self.image_label = QLabel(self)
            layout.addWidget(self.image_label)

                  self.count_label = QLabel(f'Captured Images: 0 /
{self.image_limit}', self)
            layout.addWidget(self.count_label)
```

```
            self.name_input = QLineEdit(self)
            self.name_input.setPlaceholderText('Enter your name')
            layout.addWidget(self.name_input)

            self.usn_input = QLineEdit(self)
            self.usn_input.setPlaceholderText('Enter your student number')
            layout.addWidget(self.usn_input)

            button_layout = QHBoxLayout()

            self.submit_button = QPushButton('Submit', self)
            self.submit_button.clicked.connect(self.submit_data)
            button_layout.addWidget(self.submit_button)

            self.reset_button = QPushButton('Reset', self)
            self.reset_button.clicked.connect(self.reset_form)
            button_layout.addWidget(self.reset_button)

            layout.addLayout(button_layout)

            self.camera_combo = QComboBox(self)
           self.camera_combo.addItems([f"Camera {i}" for i in self.cameras])
            self.camera_combo.currentIndexChanged.connect(self.switch_camer
a)
            layout.addWidget(self.camera_combo)

            self.setLayout(layout)

        def update_frame(self):
            ret, frame = self.cap.read()
```

```
            if ret:
                frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
                results = self.face_detection.process(frame_rgb)

                if results.detections:
                    for detection in results.detections:
                       bboxC = detection.location_data.relative_bounding_box
                        ih, iw, _ = frame.shape
                        x, y, w, h = int(bboxC.xmin * iw), int(bboxC.ymin *
ih), \
                                     int(bboxC.width * iw), int(bboxC.height
* ih)
                         cv2.rectangle(frame, (x, y), (x + w, y + h), (0,
255, 0), 2)
                        if self.image_count < self.image_limit:
                            self.capture_image(frame)

                rgb_image = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
                h, w, ch = rgb_image.shape
                bytes_per_line = ch * w
                     convert_to_Qt_format = QImage(rgb_image.data, w, h,
bytes_per_line, QImage.Format_RGB888)
               p = convert_to_Qt_format.scaled(400, 300, Qt.KeepAspectRatio)
                self.image_label.setPixmap(QPixmap.fromImage(p))

        def capture_image(self, frame):
            _, buffer = cv2.imencode('.png', frame)
            encoded_string = base64.b64encode(buffer).decode('utf-8')
            self.captured_images.append(encoded_string)
            self.image_count += 1
```

```
            self.count_label.setText(f'Captured Images: {self.image_count}
/ {self.image_limit}')

        def submit_data(self):
            name = self.name_input.text()
            usn = self.usn_input.text()

            if not name or not usn:
                print("Please enter both name and student number.")
                return

            data = {
                "name": name,
                "usn": int(usn),
                "images": self.captured_images
            }

            server_url = "http://localhost:3000/api/submit"  # Replace with
your server details

            try:
                response = requests.post(server_url, json=data)
                if response.status_code == 200:
                    print("Data sent successfully!")
                else:
                                print(f"Failed to send data. Status code:
{response.status_code}")
            except requests.exceptions.RequestException as e:
                print(f"An error occurred: {e}")

        def reset_form(self):
```

```
            self.name_input.clear()
            self.usn_input.clear()
            self.captured_images = []
            self.image_count = 0
                    self.count_label.setText(f'Captured   Images:   0   /
{self.image_limit}')

        def switch_camera(self, index):
            self.cap.release()
            self.current_camera = self.cameras[index]
            self.cap = cv2.VideoCapture(self.current_camera)

        def closeEvent(self, event):
            self.cap.release()
            self.face_detection.close()

    if __name__ == '__main__':
        app = QApplication(sys.argv)
        ex = FaceDetectionApp()
        ex.show()
        sys.exit(app.exec_())

    import sys
    import cv2
    import os
    import json
    import numpy as np
    from PyQt5.QtWidgets import QApplication, QMainWindow, QListView
    from PyQt5.QtGui import QImage, QPixmap
    from PyQt5.QtCore import QTimer, Qt, QStringListModel
    import datetime
```

```
import time
from imgbeddings import imgbeddings
from PIL import Image
from main_window_list import Ui_MainWindow_List
import mediapipe as mp
import csv

# Initialize MediaPipe Face Mesh
mp_face_mesh = mp.solutions.face_mesh
mp_drawing = mp.solutions.drawing_utils

# Load the face embeddings
with open("face_embeddings.json", "r", encoding="utf-8") as f:
    face_embeddings = json.load(f)

# Initialize imgbeddings
ibed = imgbeddings()

class MainWindow(QMainWindow, Ui_MainWindow_List):
    def __init__(self):
        super().__init__()
        self.setupUi(self)
        self.set_logo("./amaLogo.png")

        self.caps = [
            cv2.VideoCapture(0),
            cv2.VideoCapture(""),
        ]

        self.timer = QTimer(self)
        self.timer.timeout.connect(self.update_frames)
```

```
            self.timer.start(60)

            self.log_model = QStringListModel()
            self.listView.setModel(self.log_model)
            self.log_list = []
            self.logged_persons = {}
             self.finalization_threshold = 2  # 2 seconds to finalize the
detected name
              self.frame_consistency = 5  # Require consistent name for 5
frames
            self.start_time = time.time()
            self.frame_count = 0

            # Initialize MediaPipe Face Mesh
                 self.face_mesh  =  mp_face_mesh.FaceMesh(max_num_faces=10,
min_detection_confidence=0.5, min_tracking_confidence=0.5)

            # Initialize CSV logging
            self.csv_file_path = "face_detection_logs.csv"
            self.initialize_csv_log()

        def update_frames(self):
            frames = [cap.read()[1] for cap in self.caps]

            if frames[0] is not None:
                self.display_frame(frames[0], self.camera_0)
            if frames[1] is not None:
                self.display_frame(frames[1], self.camera_1)

            self.frame_count += 1
            elapsed_time = time.time() - self.start_time
```

```
            if elapsed_time > 1.0:
                fps = self.frame_count / elapsed_time
                self.setWindowTitle(f"Face Recognition - FPS: {fps:.2f}")
                self.start_time = time.time()
                self.frame_count = 0

        def display_frame(self, frame, label):
            rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
            results = self.face_mesh.process(rgb_frame)

            if results.multi_face_landmarks:
                for face_landmarks in results.multi_face_landmarks:
                    # Get bounding box from landmarks
                    h, w, _ = frame.shape
                            x_min = int(min([landmark.x for landmark in
face_landmarks.landmark]) * w)
                            x_max = int(max([landmark.x for landmark in
face_landmarks.landmark]) * w)
                            y_min = int(min([landmark.y for landmark in
face_landmarks.landmark]) * h)
                            y_max = int(max([landmark.y for landmark in
face_landmarks.landmark]) * h)

                    # Ensure the bounding box is within the frame
                    x_min, y_min = max(0, x_min), max(0, y_min)
                    x_max, y_max = min(w, x_max), min(h, y_max)

                    # Check if the face region is valid
                    if x_min < x_max and y_min < y_max:
                        # Extract face image
                        face_img = frame[y_min:y_max, x_min:x_max]
```

```
                    # Check if face_img is not empty
                    if face_img.size > 0:
                        pil_img = Image.fromarray(cv2.cvtColor(face_img,
cv2.COLOR_BGR2RGB))
                        embedding = ibed.to_embeddings(pil_img)[0]

                        name = self.recognize_face(embedding)

                        # Draw bounding box and name
                        cv2.rectangle(frame, (x_min, y_min), (x_max,
y_max), (0, 255, 0), 2)
                        cv2.rectangle(frame, (x_min, y_min-40), (x_max,
y_min), (0, 255, 0), -1)
                        cv2.putText(frame, name, (x_min, y_min-10),
cv2.FONT_HERSHEY_SIMPLEX, 0.9, (255, 255, 255), 2)

                    self.log_face_detection(label.objectName(), name)

            rgb_image = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
            h, w, ch = rgb_image.shape
            bytes_per_line = ch * w
            qt_image = QImage(rgb_image.data, w, h, bytes_per_line,
QImage.Format_RGB888)

            scaled_qt_image = qt_image.scaled(label.size(),
Qt.AspectRatioMode.KeepAspectRatio)
            label.setPixmap(QPixmap.fromImage(scaled_qt_image))

        def recognize_face(self, embedding):
            min_distance = float('inf')
```

```
            recognized_name = "unknown"

            for name, stored_embedding in face_embeddings.items():
                        distance   =   np.linalg.norm(np.array(embedding)   -
np.array(stored_embedding))
                if distance < min_distance:
                    min_distance = distance
                    recognized_name = name

            print(f"{recognized_name}: {min_distance}")
              if min_distance > 9 and min_distance < 11:  # Adjust this
threshold as needed
                recognized_name = "Loading"
            elif min_distance > 11:
                recognized_name  = "Unknown"

            return recognized_name

        def log_face_detection(self, cam_label, name):
            current_time = time.time()

            if name in self.logged_persons:
                person_data = self.logged_persons[name]

                # If the person was last seen with the same name
                 if not person_data['finalized'] and person_data['name'] ==
name:
                    time_diff = current_time - person_data['first_seen']
                    person_data['frame_count'] += 1
```

```
                    # Require the name to be consistent for at least `frame_consistency` frames
                    if time_diff >= self.finalization_threshold and person_data['frame_count'] >= self.frame_consistency:
                        self.add_log_entry(name, cam_label)
                        person_data['finalized'] = True
                else:
                    # If the name changes, reset the timer
                    person_data['first_seen'] = current_time
                    person_data['name'] = name
                    person_data['frame_count'] = 1
                    person_data['finalized'] = False

                # Update last seen time
                person_data['last_seen'] = current_time

            else:
                # New detection, store the first detection time and log it
                self.logged_persons[name] = {
                    'camera': cam_label,
                    'logged': False,
                    'last_seen': current_time,
                    'first_seen': current_time,
                    'name': name,
                    'finalized': False,
                    'frame_count': 1
                }

        def add_log_entry(self, name, cam_label):
                timestamp = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")
```

```
            log_entry = f"{timestamp} - Face detected on {cam_label}: {name}"
            self.log_list.append(log_entry)
            self.log_model.setStringList(self.log_list)
            self.listView.scrollToBottom()
            self.log_to_csv(timestamp, cam_label, name)

        def initialize_csv_log(self):
            if not os.path.exists(self.csv_file_path):
                with open(self.csv_file_path, 'w', newline='') as csvfile:
                    csv_writer = csv.writer(csvfile)
                    csv_writer.writerow(['Timestamp', 'Camera', 'Name'])

        def log_to_csv(self, timestamp, camera, name):
            with open(self.csv_file_path, 'a', newline='') as csvfile:
                csv_writer = csv.writer(csvfile)
                csv_writer.writerow([timestamp, camera, name])

        def closeEvent(self, event):
            for cap in self.caps:
                cap.release()
            self.face_mesh.close()

    def main_camera():
        app = QApplication(sys.argv)
        window = MainWindow()
        window.show()
        sys.exit(app.exec_())

    if __name__ == "__main__":
        main_camera()
```

```
import os
import cv2
import numpy as np
from imgbeddings import imgbeddings
from PIL import Image
import json
import mediapipe as mp

def train_images():
    # Initialize MediaPipe Face Mesh
    mp_face_mesh = mp.solutions.face_mesh
        face_mesh = mp_face_mesh.FaceMesh(static_image_mode=True,
max_num_faces=1, min_detection_confidence=0.5)

    # Create a dictionary to store name and embeddings
    embeddings_dict = {}

    # Initialize imgbeddings
    ibed = imgbeddings()

    # Iterate through the dataset directory
    dataset_dir = 'received_images'
    if not os.path.exists(dataset_dir):
        print(f"Error: '{dataset_dir}' directory not found.")
        exit(1)

    for name in os.listdir(dataset_dir):
        print(name)
        person_dir = os.path.join(dataset_dir, name)
        if os.path.isdir(person_dir):
```

```
                embeddings_list = []
                # Iterate through the image files in the person's directory
                for image_file in os.listdir(person_dir):
                    image_path = os.path.join(person_dir, image_file)
                     # https://stackoverflow.com/questions/43185605/how-do-
i-read-an-image-from-a-path-with-unicode-characters
                    stream = open(image_path, "rb")
                    bytes = bytearray(stream.read())
                    numpyarray = np.asarray(bytes, dtype=np.uint8)
                    image = cv2.imdecode(numpyarray, cv2.IMREAD_UNCHANGED)
                    if image is None:
                      print(f"Warning: Failed to read image '{image_path}'")
                        continue

                    # Convert the image to RGB for MediaPipe
                    rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
                    results = face_mesh.process(rgb_image)

                    if results.multi_face_landmarks:
                        for face_landmarks in results.multi_face_landmarks:
                            # Get bounding box from landmarks
                            h, w, _ = image.shape
                               x_min = int(min([landmark.x for landmark in
face_landmarks.landmark]) * w)
                               x_max = int(max([landmark.x for landmark in
face_landmarks.landmark]) * w)
                               y_min = int(min([landmark.y for landmark in
face_landmarks.landmark]) * h)
                               y_max = int(max([landmark.y for landmark in
face_landmarks.landmark]) * h)
```

```
                        # Ensure the bounding box is within the image
                        x_min, y_min = max(0, x_min), max(0, y_min)
                        x_max, y_max = min(w, x_max), min(h, y_max)

                        # Check if the face region is valid
                        if x_min < x_max and y_min < y_max:
                            # Extract face image
                            face_img = image[y_min:y_max, x_min:x_max]

                            # Check if face_img is not empty
                            if face_img.size > 0:
                                                                pil_img   =
Image.fromarray(cv2.cvtColor(face_img, cv2.COLOR_BGR2RGB))
                                embedding = ibed.to_embeddings(pil_img)[0]
                                embeddings_list.append(embedding.tolist())

                # Store the average embedding for the person
                if embeddings_list:
                    avg_embedding = np.mean(embeddings_list, axis=0)
                    embeddings_dict[name] = avg_embedding.tolist()

        # Save the embeddings to a JSON file
        with open("face_embeddings.json", "w", encoding="utf-8") as f:
            json.dump(embeddings_dict, f)

        print("Face embeddings saved successfully.")

        # Close the MediaPipe Face Mesh
        face_mesh.close()

    if __name__ == "__main__":
```

```
    train_images()


import os, json
import base64

directory = "received_images"
if not os.path.exists(directory):
    os.makedirs(directory)

dataset_dir = '../server_side/users/json'
if not os.path.exists(dataset_dir):
    print(f"Error: '{dataset_dir}' directory not found.")
    exit(1)

def read_json_file(file_path):
    with open(file_path, 'r', encoding='utf-8') as json_file:
        data = json.load(json_file)
    return data

def parse_json():
    for name in os.listdir(dataset_dir):
        data = read_json_file(os.path.join(dataset_dir, name))
        name = data.get("name")
        usn = data.get("usn")
        images = data.get("images")

        image_folder = f'{directory}/{name}'
        if not os.path.exists(image_folder):
            os.makedirs(image_folder)
```

```
            for i, img_str in enumerate(images):
                img_data = base64.b64decode(img_str)
                                file_path    =    os.path.join(image_folder,
f"{name}_{usn}_{i}.jpg")
                print(f"{name}_{usn}_{i}.jpg")
                with open(file_path, 'wb') as img_file:
                    img_file.write(img_data)

    if __name__ == "__main__":
        parse_json()
```

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LUCENA CITY, INC.

# CERTIFICATION

This is to certify that I have checked and verified the statistical result of the **CAPSTONE PROJECT** of **John Emmanuel S. Marco** and **John Kennedy H. Peña** entitled *"AMA Facial Recognition: Digital Record-logs Management System*."

This is to certify further that it conforms to generally acceptable standards for statistical treatment.

Given this 8th day of November 2024 at Davao City, Philippines.

**JOSHUA DINGDING, MSPAM, CpE**

Statistician

AMA COMPUTER COLLEGE
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# GRAMMARIAN'S CERTIFICATE

This is to certify that Mr. John Emmanuel S. Marco and John Kennedy H. Peña sought assistance from the undersigned in editing the manuscript entitled *"AMA FACIAL RECOGNITION: DIGITAL RECORD-LOGS MANAGEMENT SYSTEM"*.

This certifies further that the said thesis has been checked for grammatical and documentation-related (i.e. – referencing) concerns and is now ready for binding and for serving academic purposes.

Issued this 16th day of August , 2024 at Lucena City, Quezon Province.

Signed:

**ROSE ANN E. ECLAVEA, LPT**
Grammarian

**PRC Registration No: 1580271**

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**Curriculum Vitae**



# John Emmanuel S. Marco

Metro Manila | 0976-372-3327 | ty.johnemmanuel@gmail.com

## Experience

**SP Madrid & Associates | PITX Parañaque City** — **10/2024 - 12/2024**
**Software Developer Intern | Innovation Technology**
**Python | Typescript | React | Express | Socket-io | Hugging-face**

**ALPR** (Automatic License Plate Recognition)

- **Developed and Integrated UI for ALPR System:** Designed, implemented, and integrated a user interface into the Automatic License Plate Recognition (ALPR) project.
- **Dataset Preparation:** Labeled, cleaned, and organized datasets for AI training to ensure high-quality data for model development.

**Whisper WebGPU** (Real-Time Transcription and Communication System for Agent Training role-playing simulations)

- **Real-Time Speech-to-Text Transcriber:** Developed a real-time transcription system using Transformers.js to assist trainees in analyzing conversations during role-playing sessions.
- **WebRTC Peer-to-Peer System:** Designed and implemented a peer-to-peer connection system with WebRTC, enabling bidirectional audio transcription for realistic training scenarios.
- **Integration with Whisper:** Integrated Whisper with WebRTC to provide seamless real-time transcription, enhancing the accuracy and usability of role-playing simulations.

## Skills

- Experienced in Javascript, Typescript, Reactjs, Nextjs, Nodejs, Express, Tailwind and Python
- Can code in PHP, Laravel, Java, C++, and C#
- Worked with PostgreSql, and MySql
- Understanding in RESTful API, and Unit Testing (Jest)
- Used code versioning tools, such as Git
- Knowledgeable in using Linux

## Education

**Bachelor of Science in Computer Science**
AMA Computer College Lucena City

CERTIFICATES

- Cisco: CCNAv7: Introduction to Networks
- Cisco: Switching, Routing, and Wireless Essentials
- Cisco: CPA - Programming Essentials in C++
- Cisco: CPP - Advanced Programming in C++

CS Thesis Project
AMA Facial Recognition: Digital Record-logs Management System

- **Developed** a facial recognition system integrated with CCTV to automatically log and track individuals on campus using **Python (OpenCV)** for facial recognition.
- **Implemented secure local network storage** with **JSON** (data), **XML** (facial recognition data), and **CSV** (logs) formats.
- **Backend development** using **Node.js** for server-side scripting and **RESTful APIs** for communication between components.
- **Ensured data privacy and security** with encryption, authentication, and local network-only access.
- **Enhanced operational efficiency** by automating attendance and presence tracking, reducing manual errors.

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LUCENA CITY, INC.



# JOHN KENNEDY
# HARINGA PEÑA

## CONTACT


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145 CCF Village Brgy. Mayao Parada, Lucena City

https://gljkhp.github.io


## EDUCATION

**2009 - 2015**
MAYAO CROSSING ELEMENTARY SCHOOL

**2015 - 2019**
LUCENA CITY NATIONAL HIGH SCHOOL

**2019 - 2024**
**AMA COMPUTER COLLEGE**
- BS in Computer Science

## SKILLS

- Programmer

Java, C#, C++, Python, JS, HTML

- Microsoft Office User

Word, Powerpoint, Excel, Publisher

- Graphic Design Artist

Photoshop, Canva, Blender

## PROJECTS

1. AMA Facial Recognition: Digital Record-logs Management System (Python, Node.js)
2. NovaGPT (90k+ users, Discord Bot, Python)
3. AmongUsMenu / SickoMenu (C++)
4. meeseeks-leaderboard-api (Javascript)

## WORK EXPERIENCE

**AMA Computer College** 2024
IT Assistant

- Computer Troubleshooting
- Use of Office Apps
- Graphics Design

**AVS Marketing Corporation** 2024
Admin Officer Trainee

- Product Encoder
- Purchase and Customer Order Encoder (Tertiary and Key Accounts)
- Inventory, Sales, and Collection Report

## REFERENCE

**Ma. Jinkie B. Dizon**
AMA Computer College
Sales and Revenue Officer

**Arron Roy P. Paulo**
AVS Marketing Corporation
Operations Manager

- Word Count: 8126

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Yutika Amelia Effendi, Amila Sofiah, Niko Azhari Hidayat, Awol Seid Ebrie, Zainy Hamzah. "Predicting vulnerability for brain tumor: data-driven approach utilizing machine learning", Indonesian Journal of Electrical Engineering and Computer Science, 2024

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Sufyan bin Uzayr, "Getting the Most out of Node.js Frameworks", CRC Press, 2022

procedures frequently result in misfiled papers, data entry errors, and delays in information retrieval, all of which can seriously reduce productivity. The application of facial recognition technology offers a potential remedy for these issues. Accuracy and security of records can be greatly improved by organizations using facial recognition for digital record management. To make sure that only authorized people may access sensitive data, facial recognition systems match a person's facial traits with stored data using complex algorithms. This eliminates the need for human verification, which not only improves security but also increases record-keeping efficiency. Project Context Inefficient record-keeping is a significant challenge at AMA Computer College, where managing records manually leads to time-consuming processes and a high likelihood of errors. Communicating and managing records about student attendance, staff presence, or visitor logs becomes increasingly difficult without a centralized digital platform. This often results in miscommunication or delays in accessing crucial information, affecting the overall operational efficiency of the institution. Accessibility issues with manual record management make it difficult for administrators to swiftly retrieve and validate records. This inefficiency raises the possibility of data loss or the misplacement of crucial papers in addition to impeding administrative productivity. Furthermore, the lack of automated tracking technologies makes it challenging to keep an eye on and confirm people's whereabouts and actions on campus, which may jeopardize accountability and security. The implementation of the AMA Facial Recognition: Digital Record-logs Management System aims to address these challenges by leveraging advanced facial recognition technology to automate the identification and logging process. This system will ensure accurate and secure digital records of all individuals passing in front of CCTV cameras on campus. By transitioning from manual to automated processes, the system will reduce administrative burdens, minimize errors, and enhance the overall security of record management. Furthermore, the digital platform will facilitate real-time updates and easy access to information, thereby improving the efficiency and effectiveness of the college’s administrative operations. Purpose and Description To improve campus security and expedite record-keeping at AMA Computer College, the AMA Facial Recognition: Digital Record-Logs Management System is a cutting- edge solution. With the use of current CCTV equipment and cutting-edge facial recognition technology, this system automatically logs and maintains records of people entering monitored areas. The main job is to scan CCTV camera video feeds in real- time, using advanced facial recognition algorithms to identify people and produce digital records that are timestamped to reflect their presence on campus. Importantly, all critical data is kept inside the safe digital perimeter of AMA College because this solution only works within its local network. The system's main output is an Excel file, which acts as the main location for all of the data gathered and provides a comfortable and easily accessible. This system is a multi-faceted data management approach that optimizes storage and accessibility. The system utilizes a JSON-based database to store comprehensive data, providing flexibility and efficiency in data handling. Facial recognition data is stored in XML files, allowing for structured storage of complex facial features and metadata. For logging purposes, the system generates CSV files, offering a simple and widely compatible format for recording system activities and events. Authorized personnel can access and manipulate these files only when connected to the college's local network, significantly enhancing data security by preventing external access. This multi-format approach allows for easy searching, sorting, and analysis of the digital record-logs, enabling staff to quickly retrieve information as needed. Real-time monitoring capabilities are integrated into the system, with the ability to update the JSON database, XML files, and CSV logs instantly as new data is captured. This feature provides immediate insights into campus activity, allowing for prompt responses to any security concerns or anomalies detected by the facial recognition system. Security and privacy considerations are paramount in the design of the AMA Facial Recognition: Digital Record-logs Management System. The use of a JSON-based database for comprehensive data storage, XML files for facial recognition data, and CSV files for logs, combined with the system's confinement to the local network, adds multiple layers of security, significantly reducing the risk of unauthorized access or data breaches. Access to these files is strictly controlled, with robust encryption and authentication mechanisms ensuring that only authorized personnel can view or modify the data. By automating the process of detecting and logging individuals on campus into this multi-format data storage system, this solution not only enhances security measures but also significantly reduces the administrative burden on staff. The structured nature of JSON, XML, and CSV formats ensures that the system remains efficient and easily maintainable, while still providing powerful data management capabilities. The JSON database offers flexibility for complex data structures, XML provides a standardized format for facial recognition data, and CSV logs offer straightforward activity tracking. Ultimately, the AMA Facial
Recognition: Digital Record-logs Management System represents

**a significant step forward in leveraging technology to** 17

create a safer, more efficient educational environment, all while maintaining strict control over data access and security through its local network implementation and multi-format data storage solution.

**Objective of the Study The main** objective **of this research is to** develop **a** 10

facial recognition-based digital record-logs management system that revolutionizes campus security and operations at AMA Computer College. This system aims to offer enhanced security, efficiency, and better engagement. Specifically,

**this study aims to**: 1. **Develop a** system **that** efficiently organizes **data** 15

and tracks individuals passing through CCTV-monitored areas using facial recognition software. 2. Create a comprehensive system capable of producing all necessary reports for security and administrative purposes. 3. Implement a secure, local network-based solution that stores facial recognition data in XML format, general data in a JSON-based database, and activity logs in CSV files. Significant of the Study An important development in AMA Computer College's campus security and operational effectiveness is the AMA Facial Recognition: Digital Record-logs Management System. This cutting-edge solution automates the process of tracking people on campus using CCTV cameras, which significantly improves daily operations. Campus safety is given priority while operational efficiency is ensured by simplifying security protocols and lowering manual labor. The system's real-time monitoring and alert capabilities enable quick responses to security concerns, fostering a secure environment for students and staff. Moreover, its robust data management approach, utilizing JSON, XML, and CSV formats within a confined local network, demonstrates a strong commitment to data security and privacy protection. The implementation of this facial recognition system brings numerous benefits to administrative efficiency and technological integration. It improves overall institutional administration by helping with a variety of administrative duties, such as monitoring the population flow on campus and producing thorough reports. By adding state-of-the-art facial recognition technology to its current CCTV systems, AMA Computer College puts itself at the forefront of advancements in campus security. This project offers significant chances for staff training and development in cutting-edge security technologies in addition to improving the college's IT infrastructure. By demonstrating the institution's commitment to safety and responsible technology use, the effective setup of this system can greatly increase community trust while upholding privacy and ethical standards. Project Scope and Limitations Scope of the Study The project will create a facial recognition system that is connected to the campus's current CCTV cameras at AMA Computer College. This technology will provide a complete solution for campus security and management by capturing and processing the face data of people moving through regions that are being monitored. In order to guarantee effective data handling and storage, the project will employ a multifaceted strategy: activity logs will be stored in CSV files, facial recognition data will be stored in XML files, and general data will be stored in a JSON-based database. The entire system will function solely within the college's local network, avoiding any unauthorised external access, in order to preserve the highest level of security and data protection. Delimitation of the Study The AMA Computer

College campus's areas that are already under surveillance by CCTV cameras are the only places where the facial recognition technology can operate. Depending on ambient circumstances including lighting, camera angles, and other obstructions, its accuracy may change. Off-campus and remote access features are purposefully absent from the system due to security concerns. The processing power and storage resources that the college's IT infrastructure allots determine the system's performance and capability. Chapter II Literature Review on Related Works Foreign Literature In recent years, numerous face recognition-based attendance management systems have been introduced to improve student performance in various organizations.

> **The human face is a sophisticated multidimensional structure that can convey** significant **information about the individual, including expression**, feelings, and **facial features** 3

. Face recognition technology was first developed by Woodrow Wilson Bledsoe in 1966, utilizing a facial database to match faces in digitized photos or videos. The concept of face recognition emerged to enable computers to efficiently and accurately identify human faces. According to Pawaskar and Chavan [1], numerous algorithms have been developed to improve the performance of face recognition. Effectively and efficiently analyzing

> **facial information is a** complex **task that requires** significant **time and** 3

effort. Kowsalya [2] mentions that various

> **facial recognition-based algorithms for** automated **attendance management** have recently **been proposed**, effectively **implemented, and** 3

put into practice. The system uses Principal Component Analysis (PCA) technology to detect each student's face, which is crucial for implementing the attendance system. This technique reduces the number of variables involved in face recognition [3]. Despite much of the literature focusing on standalone systems, web-based systems are preferred due to their benefits such as quick

> **data access, scalability, responsiveness, interactivity, remote management, and centralized data** 1

. Nugroho and Fajar [4] note that user perspectives

> **are influenced by factors** like **optimism (a positive view of technology's ability to enhance work efficiency) and innovativeness (enthusiasm for** 1

registered using facial recognition. However, this system may still register students who leave the classroom, potentially leading to errors in attendance data. Another method involves using a camera within the classroom, capturing a picture of the classroom throughout the lesson. The proposed system addresses these limitations, offering the benefits of flexible usage, no wasted time, and easy accessibility. Local Literature Classrooms and corporate enterprises are just two examples of contexts where attendance monitoring, an application of people counting, proves useful. Despite its vast potential, automated people, counting and attendance monitoring systems are relatively scarce in the Philippines. For example, manual hand counting is used in parks, terminals, supermarkets, and shopping centers. Security staff are provided with portable

> **counting devices to** track **the number of** individuals **entering** terminals or **buildings. However, manual attendance methods are labor-intensive**, time-consuming, **and costly** 7

[1]. Foreign Studies Numerous

> **studies in the literature** primarily **focus on** the **technical aspects of automated attendance recording** and algorithm accuracy, **often overlooking** important **post-processing of attendance data** needed **for** managing absenteeism, **auditing**, monitoring, **and generating reports** 1

. Additionally, there is limited research on user preferences, adoption,

> **and satisfaction with these technologies**. This project aims **to address this gap** 1

by developing

> **a web-based attendance management system** utilizing QR code **and** facial recognition **technology**. The **system** 1

provides a cost-effective solution by reducing the need for extensive hardware installation and maintenance through its configurable options, while emphasizing the importance of setting up technology infrastructure in

> **educational institutions, considering factors** like **internet connectivity, hardware, platforms, and staff** 1

**Raspberry Pi captures an image**, the **Local Binary Pattern** (LBP) **algorithm is applied to** match **it**

against images stored in the database. If a match is found,

**a servo motor opens the door for the student. This system** achieved **an accuracy of 95% with a dataset of 11 images** 2

. Damale and Pathak [3] mention

**several existing systems closely related to the idea of** recording attendance **in class using facial recognition techniques and algorithms** 5

. Their literature review of these systems shows their effectiveness. The proposed case study is based on

**sources related to facial recognition and image processing** 5

. Chinimilli et al. [4] note that these systems use cameras

**to capture the faces of all students**, either **simultaneously or one** 1

at a time. Facial recognition is increasingly used for student attendance tracking in educational institutions [5]. Kamencay [6] highlights that facial recognition improves the attendance system, eliminates errors in manual recording, provides automatic and reliable attendance solutions, enhances privacy and security, prevents false attendance, and generates regular attendance reports. Gornale and Kiran [7] pointed out that facial recognition technology saves time for educators and students, while preventing fake attendance. The process involves capturing student videos, converting them into frames, matching them to a database, and marking attendance based on recognition. Local Studies The traditional method of determining attendance in a class has limitations. Automated attendance methods were introduced to address these issues. Rollon et al. [1] successfully developed a web application useful for both professors and students. The product quality is determined by factors like functional appropriateness, reliability, performance efficiency, and usability. This technology enables accurate and automatic attendance tracking. The OpenCV Real-time Face Recognition Attendance System is advantageous as it saves time for teachers and users in recording attendance. By capturing an image recognized by OpenCV, the system automatically records the names registered in the image and can convert the data into an Excel file for documentation purposes [2]. A commonly used

**face detection** approach, **Histogram of Oriented Gradients (HOG) with** 4

**distance from the camera** on **facial detection and recognition. The obtained attendance** data is **uploaded to a database with authentication** 4

. The system demonstrated an accuracy of 95.65% and was capable of detecting and recognizing up to 37 students. To improve the system's accuracy, a brightness level of at least 217.39 lux is recommended. Furthermore, the student's distance from the camera did not affect the system's accuracy. Arceo et al. [3] suggested that for optimal recognition, students should face the camera squarely. Synthesis Traditional campus security systems frequently rely on human monitoring and record- keeping, but the AMA Facial Recognition: Digital Record-Logs Management System provides a holistic solution for improving campus safety and operational efficiency. Educational institutions must safeguard the safety of their students and faculty while also properly managing campus activities. As a result, a college-specific automated facial recognition and record-logging system is required. AMA Computer College may utilise this technology to handle campus access, attendance tracking, security monitoring, and individual profiles more effectively. Information and communication technology is rapidly evolving, and educational institutions must employ cutting-edge solutions to keep up with the newest developments in the area. This technical advancement is seen in the integration of facial recognition with existing CCTV systems, which are presently one of the most extensively utilised surveillance mediums. Due to the different issues that occur in campus management, which affect not only administrative obligations but also student and staff safety, educational institutions in this digital era must constantly be prepared and at the forefront of technology adoption. Technical Background To implement the Facial Recognition: Digital Entry/Exit Record-logs Management System, the following technical background and tools are required: Python (OpenCV):

**OpenCV (Open Source Computer Vision) is an open-source computer vision library** that allows **for** 9

real-time image processing. OpenCV will be used for facial recognition, including: • Face detection: Using pre-trained models such as Haar cascades to detect faces in images. • Facial feature extraction: Using various algorithms, such as Local Binary Pattern Histograms (LBPH) or Convolutional Neural Networks (CNN), to extract unique facial features. • Training and recognition: Using

**machine learning algorithms to train the** 5

system to recognize individual faces and match them with stored records.

**Node.js: Node.js is a JavaScript runtime environment that allows** for **server-side** scripting and development **of** 8

**web applications. Node.js** will be **used for** 18

**Code: Visual Studio Code (VS Code) is** an open-source, **cross-platform** 6 **code editor that supports a wide range of programming languages**
, including Python and JavaScript. VS Code will be used for: • Code editing: Writing, debugging, and testing the code for both the Python and Node.js components of the system. • Version control: Managing code changes and collaborating with other developers using Git integration. • Extensions: Utilizing various extensions to enhance the development experience, such as debuggers, linters, and code formatters. Interoperability and Integration: The system will be designed to ensure smooth connectivity between the different software programs utilized, such as the Python OpenCV library, Node.js server, and user interface. APIs and interoperability standards, such as RESTful APIs and JSON, will be employed to facilitate communication between these components and ensure that data is easily accessible and transferable. Chapter III Research Methodology Software Design, Products and/or Processes The Facial Recognition Digital Record-Logs Management System is a cutting-edge software solution that automates and streamlines the monitoring and logging of entry/exit activities, with a focus on data security and user privacy. This chapter describes the system's architecture, components, and essential features, giving a thorough overview of its design and execution. The system combines cutting-edge facial recognition technology with safe data management methods to provide a strong and efficient digital record-keeping solution. By automating the entry/exit logging process, the system hopes to improve security, minimise manual labour, and offer reliable, real-time statistics on facility access. Key components of the system include: Data Input and Storage: A user-friendly platform for inputting and securely storing user data, such as names, IDs, and high-resolution face photos. This information is stored on a dedicated server to provide the utmost security and accessibility. Facial Recognition Training: An innovative system that analyses and maps each user's unique facial traits from stored photographs. This is the foundation for the system's recognising capabilities. Real-Time Entry/Exit Logging: Use current surveillance technology to collect photographs of people at entry/exit points. The system uses real-time face recognition algorithms to compare these photographs to previously recorded data, registering entrances and exits depending on successful matches. The following sections will delve into the cost-benefit analysis, requirement specifications, and technical background necessary for implementing this state-of-the- art system.

**By thoroughly** examining **these aspects, we aim to provide a** clear 11
roadmap for **the** development **and** deployment of **the**

Facial Recognition Digital Record-Logs Management System, ensuring its effectiveness, security, and compliance with relevant regulations. System Architecture Figure 3.1 The Facial Recognition Digital RecordLogs Management System employs a multi- tiered architecture to ensure scalability, security, and efficient data processing: Client Layer: · User Interface: Web-based interface for administrators to manage user data and access logs Application Layer: · Authentication Server: Handles user authentication and access control · Facial Recognition Engine: Processes images and performs facial recognition · Logging Service: Records entry/exit events and manages historical data Data Layer: · User Database: Stores user profiles, credentials, and facial data · Log Database: Maintains a secure record of all entry/exit events · Image Storage: Securely stores captured images for processing and auditing Integration Layer: · Camera Integration Module: Interfaces with existing surveillance cameras · API Gateway: Provides secure access to system functionalities for external applications The system uses RESTful APIs for communication between layers, ensuring loose coupling and easier maintenance. Conceptual Design Figure 3.2 By The facial recognition digital record-logs management system aims to automate and streamline the monitoring and logging, while ensuring data security and privacy. The system will consist of the following components: Data Input and Storage: · A platform for entering and storing user data, including:+ Name+ ID+ High-resolution facial image · A dedicated server for securely storing the collected data Facial Recognition Training: · An algorithm for analyzing and mapping unique facial features for each user based on the stored images Real-Time Entry/Exit Logging: · Integration with existing surveillance cameras for capturing images of individuals approaching the designated entry or exit points · Real-time facial recognition algorithms for comparing captured images with stored facial data · Logging based on successful facial recognition

Cost Benefit Analysis Benefits: · Improved security and accuracy in tracking students · Reduced labor costs associated with manual log management · Enhanced data privacy and security through robust compliance measures · Scalable and adaptable solution for various entry/exit points and user volumes Costs: · Initial investment in hardware (e.g., cameras, servers) and software development · Ongoing maintenance and upgrades to ensure system performance and security · Training and support for system administrators and users · Compliance with data protection laws and regulations Requirement Analysis Functional requirements Data Entry and Storage: • The system should include an easy-to-use interface for inputting and saving user information. • Users' data should be securely stored on a dedicated server. • Facial Recognition Training: • The system should properly analyse each user's unique face traits based on recorded photos. Real-time Entry/Exit Logging • Integrate with existing surveillance cameras to collect photos of anyone approaching designated entry/exit points. • The system should perform real-time facial recognition to match collected photographs to stored data. • log successful facial recognition. Non-functional requirements Compliance and Security Measures: • The system should follow data protection rules and regulations. • Enforce strong security measures to safeguard user data and privacy. Performance: • The technology should enable rapid and accurate facial recognition logging. Scalability: • The system should be scalable to handle different points and user volumes. Usability: • The system must be user-friendly for both administrators and users. Maintainability: • The system should be simple to maintain and upgrade for best performance and security. System Flow Chart Figure 3.3 Agile Methodology Figure 3.4 This stage of agile project management identifies the project's requirements and goals. The requirement phase is when you develop solutions to meet the project's design and finalize and document all of your decisions. The project management process's implementation phase is where your plan and design are put into action to get the desired result. Testing validates that the product generated during the implementation stage meets all of the project's requirements. During the maintenance phase, minor changes are done to improve the product that was developed during implementation. System Flow

**Figure 3.5 Block Diagram Figure 3.6** **16**

Development and Testing Development Approach: · Agile methodology with 2-week sprints · Version control using Git with feature branching strategy ·

**Continuous Integration/Continuous Deployment (CI/CD) pipeline for 13 automated testing and deployment**

Testing Strategy: · Unit Testing: For individual components using frameworks like Jest for JavaScript and pytest for Python · Integration Testing: To ensure proper communication between system components · Performance Testing: Using tools like Apache JMeter to simulate high user loads · Security Testing:
Including penetration testing and vulnerability assessments Key Testing Areas: · Facial Recognition
Accuracy: Extensive testing with diverse datasets to ensure high recognition rates and low false positives · Data Privacy and Security: Rigorous testing of encryption, access controls, and compliance with data protection regulations · System Performance: Testing system responsiveness under various load conditions
· User Interface: Usability testing to ensure intuitive operation for both administrators and end-users
Description of the Prototype The prototype of the Facial Recognition Digital Record-Logs Management System demonstrates core functionalities: Facial Recognition: · Integration with a test camera setup · Realtime facial recognition for a limited user set · Logging of successful recognitions Access Logging: · Real-time logging of events · Basic reporting interface for access logs Security Features: · Basic encryption of stored data · Role-based access control for system functions The prototype serves as a proof-of-concept, showcasing the system's potential while allowing for user feedback and further refinement. It implements core facial recognition and logging functionalities, providing a foundation for the full-scale system development. This early version enables stakeholders to interact with the system, validate its usefulness, and identify areas for improvement before proceeding to full

development. Chapter IV Project Evaluation Pre-Implementation Survey Results The pre-implementation survey for the AMA Facial Recognition: Digital Record-logs Management System sought to gauge expectations, usability preferences, and possible issues among intended users and stakeholders. This poll gave useful insights regarding the system's expected success, with an emphasis on elements such as efficiency, data security, and simplicity of access. The results demonstrated a significant degree of interest in the system, notably its capacity to automate attendance and record-keeping using facial recognition technology. Respondents emphasised the importance of rapid and accurate log-ins, which they feel would improve operational efficiency and minimise time spent manually entering data. The capacity to check attendance records in real time and manage access control were also regarded as useful features, particularly for establishing a secure atmosphere on campus. Feedback indicated a preference for a straightforward user interface that can accommodate consumers of varied degrees of technological knowledge. Respondents also emphasised a desire for dependable performance, with ideas for reducing downtime and improving facial recognition accuracy in a variety of lighting situations. Concerns about data security and privacy were expressed, with recommendations for stringent data protection measures and adherence to data privacy legislation. Overall, the poll findings revealed that the AMA Facial Recognition: Digital Record- Logs Management System was well received, with expectations centred on operational efficiency, user-friendly design, and strong data security. This input will be critical in directing the final development phases and ensuring that the system meets the needs and expectations of its users. Table 7 – Evaluation of the Current System's Functionality IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The facial recognition system would be a valuable resource for managing entry/exit logs.1.68STRONGLY DISAGREE2. Users would find it helpful to have information about their entry/exit history readily available.1.64STRONGLY DISAGREE3. The system will provide up-to- date and accurate information about facility usage.1.56STRONGLY DISAGREEOver- all $\bar{x}$1.63STRONGLY DISAGREE Table 7, titled "Evaluation of Current System Functionality," provides an assessment of respondents' perceptions regarding the effectiveness and adequacy of the existing system. The results show an overall weighted mean of 1.63, with a descriptive equivalent of "strongly disagree." This rating reflects a clear consensus among respondents that the current system does not meet the functional needs or expectations required for efficient record-keeping and attendance monitoring. The "strongly disagree" descriptive equivalent associated with this weighted mean underscores significant dissatisfaction with the current system's functionality, revealing a critical need for a more advanced solution. The consistent feedback from respondents signals a strong demand for improvements, particularly in areas such as accuracy, efficiency, and security of log management, which are currently perceived as insufficient. The findings from Table 7 strongly indicate a need for the implementation of the AMA Facial Recognition: Digital Record-logs Management System. This system is expected to address the gaps highlighted in the survey, offering automated facial recognition technology for efficient, accurate, and secure record-keeping. The strong disapproval reflected in these results emphasizes the importance of moving forward with this system to better meet users' needs and enhance operational efficiency. . Table 8: Evaluation of Current System's A.I Integration IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. An AI system could effectively analyze usage patterns and provide valuable insights.1.36STRONGLY DISAGREE2. Users would trust an AI-powered system to provide accurate and up-to- date access information.1.88DISAGREE3. Integrating AI into the facial recognition system is technically feasible.1.56STRONGLY DISAGREEOver-all $\bar{x}$1.60STRONGLY DISAGREE Table 8, titled "Evaluation of Current System's A.I. Integration," presents an analysis of respondents' perceptions regarding the functionality and adequacy of artificial intelligence within the current system. The data reveals an overall weighted mean of 1.60, corresponding to a descriptive equivalent of "strongly disagree." This indicates a consensus among respondents that the current A.I. integration fails to meet their expectations and functional needs. The "strongly disagree" descriptive equivalent associated with this weighted mean reflects significant dissatisfaction with the current system's A.I. capabilities, underscoring the urgency for a more effective solution. Respondents consistently expressed a need for enhanced A.I. integration to improve accuracy, efficiency, and real-time capabilities, as the existing system is perceived as inadequate in these areas. These results highlight a critical need for the AMA Facial Recognition: Digital Record- logs Management System, which promises advanced A.I. integration tailored to meet user needs for accurate and efficient record management. The strong disapproval reflected in Table 8 reinforces the demand for a more robust system that leverages A.I. effectively, making a compelling case for the implementation of this proposed system. Table 9: Evaluation of the Current System's Image Processing Capabilities IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The quality of facial recognition images will be sufficient for accurate identification.1.48STRONGLY DISAGREE2. Implementing high-quality image capture would be costeffective

for the organization.1.28STRONGLY DISAGREE3. Advanced imaging techniques would not negatively impact system performance or speed.1.04STRONGLY DISAGREEOver-all x̄1.27STRONGLY DISAGREE Table 9, titled “Evaluation of the Current System’s Image Processing Capabilities,” presents an analysis of respondents' perceptions regarding the adequacy of the current system’s ability to process images effectively. The data reveals an overall weighted mean of 1.27, corresponding to a descriptive equivalent of "strongly disagree." This score reflects a strong consensus among respondents that the current image processing capabilities are inadequate and fall short of their needs and expectations. The "strongly disagree" rating associated with this weighted mean highlights significant dissatisfaction with the system’s image processing functionality, underscoring the urgent need for an enhanced solution. Respondents consistently expressed concerns about the reliability and accuracy of the current system in handling image data, signaling a strong need for an upgrade in technology to meet the demands of efficient record-keeping and accurate data processing. The results in Table 9 emphasize the necessity of the AMA Facial Recognition: Digital Record-logs Management System. This proposed system, with advanced image processing capabilities, is anticipated to bridge the critical gaps identified in the current system, providing precise and dependable image analysis and record management. The strong disapproval reflected in these findings highlights a substantial demand for a more robust solution, making a compelling case for the implementation of the AMA Facial Recognition system to comprehensively address the needs of its users. Table 10: Evaluation of the Current System’s Performance IndicatorsWEIGHTED MEAN (x̄)DESCRIPTIVE EQUIVALENT1. The system will be easy to maintain and update.1.20STRONGLY DISAGREE2. The system will be reliable and have minimal downtime.1.17STRONGLY DISAGREE3. The system will be able to handle high traffic loads without performance issues.1.56STRONGLY DISAGREEOver-all x̄1.31STRONGLY DISAGREE Table 10, titled “Evaluation of the Current System’s Performance,” presents an analysis of respondents' perceptions regarding the overall performance of the current system. The data reveals an overall weighted mean of 1.31, corresponding to a descriptive equivalent of "strongly disagree." This indicates a strong consensus among respondents that the current system fails to meet their performance expectations, highlighting its inefficiency and inability to function effectively according to user needs. The "strongly disagree" rating associated with this weighted mean underscores significant dissatisfaction with the current system’s performance, signaling a critical need for improvement. Respondents consistently expressed concerns about the system’s reliability, speed, and overall effectiveness in managing digital record logs, further emphasizing the gap in meeting necessary operational standards. The findings from Table 10 make a compelling case for the AMA Facial Recognition: Digital Record-logs Management System, which is designed to offer a substantial performance upgrade over the existing system. This proposed system promises enhanced efficiency, accuracy, and functionality, addressing the critical shortcomings identified in the current system. The strong disapproval reflected in these results reinforces the need for a more reliable and effective solution to meet user demands and expectations. Table 11: Evaluation of the Current System’s Security IndicatorsWEIGHTED MEAN (x̄)DESCRIPTIVE EQUIVALENT1. Confident enough the system will be secure and protect user data.1.28STRONGLY DISAGREE2. Can system be resistant to cyberattacks and data breaches.1.20STRONGLY DISAGREE3. User authentication processes will be secure and reliable.1.68STRONGLY DISAGREEOver-all x̄1.39STRONGLY DISAGREE Table 11, titled “Evaluation of the Current System’s Security,” presents an analysis of respondents' perceptions regarding the security capabilities of the current system. The data reveals an overall weighted mean of 1.31, corresponding to a descriptive equivalent of "strongly disagree." This score reflects a strong consensus among respondents that the current system's security measures are insufficient and fail to meet the necessary standards for protecting sensitive data. The "strongly disagree" descriptive equivalent associated with this weighted mean indicates a significant perception among respondents that the current system lacks the required security features, highlighting concerns over data integrity, user privacy, and overall system protection. This outcome signals a critical gap in the current system's ability to safeguard sensitive information, especially in an era where security threats are increasingly sophisticated. The results in Table 11 emphasize the urgent need for the AMA Facial Recognition: Digital Record-logs Management System, which is designed to provide a higher level of security. The new system incorporates advanced security protocols, including facial recognition technology, to ensure robust data protection, prevent unauthorized access, and maintain the confidentiality and accuracy of the record logs. The strong disapproval reflected in these results further reinforces the need to transition to a more secure, reliable system that can effectively address the security challenges identified in the current system. Post-Implementation Survey Results The post-implementation survey for the AMA Facial Recognition: Digital Record-logs Management System was conducted to evaluate the

system's ability to meet the expectations of its users and stakeholders following deployment. The survey aimed to assess the system's performance, user experience, security features, and overall satisfaction with its usefulness in real-world settings. Users provided an extremely favorable response, particularly regarding the system's effectiveness in automating attendance and record-keeping activities. Respondents noted considerable time savings due to the accurate and speedy facial recognition technology, which expedited log-ins and reduced the need for manual data entry. Real- time attendance monitoring and access control capabilities were highly valued by users, who stated that these features enhanced overall security and helped maintain a more organized record-keeping system within the campus environment. The system's user interface was also well-received, with users finding it straightforward and easy to use, especially for those with limited technical knowledge. Most users were satisfied with the system's functionality, reporting no downtime and finding the facial recognition to be accurate across a wide range of lighting conditions. The system's reliability in maintaining records and managing access control was also praised, fostering a sense of trust among users. Concerns raised about data security and privacy during the pre-implementation phase were addressed in the final system design, and respondents expressed satisfaction with the security measures implemented. Many pointed out that the system adhered to data protection standards and privacy regulations, which reassured them about the confidentiality and integrity of their personal information. Overall, the survey findings showed strong approval for the AMA Facial Recognition: Digital Record-logs Management System. The system met users' expectations in terms of efficiency, convenience, and security. Positive feedback and recommendations will be crucial for future system upgrades, ensuring that the system continues to meet the evolving needs of its users. The system's successful implementation demonstrates its potential to enhance operational efficiency and security on campus. Table 12: Evaluation of the Implemented System’s Functionality IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The facial recognition system has improved the efficiency of managing entry/exit logs.3.96AGREE2. The automated digital record-keeping has met the facility's access management needs.4.72STRONGLY AGREE3. Users find the readily available information about their entry/exit history helpful.4.28STRONGLY AGREEOver-all $\bar{x}$4.32STRONGLY AGREE Table 12, titled “Evaluation of the Implemented System’s Functionality,” presents an analysis of respondent perceptions regarding the performance and functionality of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.32, corresponding to a descriptive equivalent of "strongly agree." This suggests a high level of satisfaction among respondents, indicating that the system has effectively met their expectations in terms of functionality. The "strongly agree" descriptive equivalent associated with the overall weighted mean reflects a consensus among users that the system is fulfilling its intended purpose. Respondents noted that the system’s facial recognition feature operates accurately and efficiently, streamlining attendance and record-keeping tasks while enhancing operational productivity. The positive responses underscore the system’s success in providing a reliable and userfriendly platform, which has greatly contributed to effective record management and secure access control. A closer examination of the individual responses in Table 12 further reinforces this satisfaction. Users consistently expressed that the system met their expectations, with minimal downtime and robust functionality under various conditions. The system’s intuitive interface and dependable performance in maintaining records demonstrate its value in campus operations, providing a well-rounded solution that addresses the needs of both security and convenience. In summary, the findings from Table 12 demonstrate that the AMA Facial Recognition: Digital Record-logs Management System has not only met but exceeded user expectations in terms of functionality. The "strongly agree" sentiment among respondents highlights the system’s effectiveness, affirming that it is operating as intended and delivering substantial benefits to the campus community. Table 13: Evaluation of Implemented System’s A.I Integration IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The AI component effectively manages complex queries about access patterns and usage.3.92AGREE2. The AI system provides valuable insights through its analysis of usage patterns.4.36STRONGLY AGREE3. The integration of AI into the facial recognition system has been smooth and effective.4.24STRONGLY AGREEOver-all $\bar{x}$4.17AGREE Table 13, titled “Evaluation of Implemented System’s A.I. Integration,” presents an analysis of respondent perceptions regarding the effectiveness of the A.I. integration in the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.17, corresponding to a descriptive equivalent of "agree." This indicates a positive consensus among respondents, suggesting that the A.I. capabilities of the system have generally met user expectations. The "agree" descriptive equivalent associated with the overall weighted mean reflects satisfaction with the system's A.I. features, specifically its facial recognition functionality and its role in enhancing efficiency and accuracy in record-keeping. Respondents noted that the A.I. integration has successfully streamlined

attendance management, facilitated secure access control, and improved overall operational efficiency. The system's effective use of A.I. to automate these processes highlights its alignment with user needs and expectations, further validating the system's performance. A closer examination of the individual responses in Table 13 further reinforces this satisfaction. Users consistently affirmed that the A.I. component is reliable and contributes meaningfully to the system's overall functionality. The responses indicate that the A.I. integration works well under various operational conditions, providing consistent and accurate results. This positive feedback reinforces the system's effectiveness as an intelligent, efficient solution for digital record-logs management on campus. In summary, the findings from Table 13 indicate that the AMA Facial Recognition: Digital Record-logs Management System has successfully met user expectations in terms of A.I. integration. The "agree" sentiment among respondents confirms the system's capability to perform as intended, delivering meaningful advantages in operational efficiency and security through its A.I.-driven features. Table 14: Evaluation of the Implemented System's Image Processing Capabilities IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The quality of facial recognition images is sufficient for accurate identification.3.96AGREE2. The implemented imaging techniques have significantly improved the system's accuracy.4.28STRONGLY AGREE3. The system demonstrates sufficient processing power to handle complex image analysis.4.32STRONGLY AGREEOver-all $\bar{x}$4.19AGREE Table 14, titled "Evaluation of the Implemented System's Image Processing Capabilities," presents an analysis of respondent perceptions regarding the effectiveness of image processing within the AMA Facial Recognition: Digital Record- logs Management System. The data reveals an overall weighted mean of 4.19, corresponding to a descriptive equivalent of "agree." This reflects a positive consensus among respondents, suggesting that the image processing capabilities of the system have largely met user expectations. The "agree" descriptive equivalent associated with the overall weighted mean indicates satisfaction with the system's ability to accurately and efficiently process images, which is critical for reliable facial recognition and secure record-keeping. Respondents noted that the image-processing component enhances the system's performance, enabling precise facial identification that supports accurate attendance logging and secure access control on campus. This outcome affirms that the system's image processing functions are working well and contributing to its overall effectiveness. A closer examination of the individual responses in Table 14 further emphasizes this positive reception. Users consistently acknowledged that the image-processing feature is reliable and responsive under different conditions, providing the accuracy and stability crucial for the system's intended purpose. This consistent positive feedback underscores the system's success in delivering on its functional promises, particularly in terms of image processing quality. In summary, the findings from Table 14 demonstrate that the AMA Facial Recognition: Digital Record-logs Management System has effectively met user expectations in terms of image processing capabilities. The "agree" sentiment among respondents validates the system's role in facilitating efficient and accurate digital record management through its robust image processing technology. Table 15: Evaluation of the Implemented System's Performance IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The system loads quickly and efficiently in real-world usage.4.28STRONGLY AGREE2. Maintenance and updates of the system have been straightforward and manageable.3.84AGREE3. The system has integrated seamlessly with other existing security and management platforms.4.48STRONGLY AGREEOver-all $\bar{x}$4.20AGREE Table 15, titled "Evaluation of the Implemented System's Performance," presents an analysis of respondent perceptions regarding the effectiveness and responsiveness of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.20, corresponding to a descriptive equivalent of "agree." This indicates a favorable response among respondents, suggesting that the system's performance has met their expectations. The "agree" descriptive equivalent associated with the overall weighted mean reflects positive user satisfaction with the system's speed, reliability, and ability to handle tasks such as automated log-ins and attendance tracking efficiently. Respondents reported that the system operates smoothly, with minimal interruptions, and fulfills its intended role in supporting accurate record management and secure campus access. This result confirms that the system's performance aligns well with its designed functionalities and user requirements. Further analysis of the responses in Table 15 reinforces the perception that the system performs reliably and consistently under typical usage scenarios. Users noted its efficient processing speed and stability, which contribute to seamless operation and an enhanced user experience. These factors underscore the system's effectiveness and support its role in promoting operational efficiency. In summary, the findings from Table 15 validate that the AMA Facial Recognition: Digital Record-logs Management System has effectively met performance expectations. The "agree" sentiment among respondents demonstrates overall satisfaction with the system's functionality and reinforces its suitability for improving digital

record keeping and campus security. Table 16: Evaluation of the Implemented System's Security IndicatorsWEIGHTED MEAN ($\bar{x}$)DESCRIPTIVE EQUIVALENT1. The facial recognition system has maintained robust security and protected user data.4.28STRONGLY AGREE2. The system has successfully resisted any potential spoofing or false identification attempts.4.44STRONGLY AGREE3. The system's mechanisms for detecting and preventing security threats have been effective.4.16AGREEOver-all $\bar{x}$4.29STRONGLY AGREE Table 16, titled "Evaluation of the Implemented System's Security," presents an analysis of respondent perceptions regarding the security features of the AMA Facial Recognition: Digital Record-logs Management System. The data reveals an overall weighted mean of 4.20, corresponding to a descriptive equivalent of "agree." This suggests a positive consensus among respondents, indicating that the system's security measures align with user expectations. The "agree" descriptive equivalent associated with the overall weighted mean reflects user confidence in the system's ability to safeguard data and manage access securely. Respondents noted the effectiveness of the security protocols in place, including access control and secure data storage, which contribute to a safe and controlled environment for campus access and attendance tracking. These findings underscore the system's success in meeting security requirements and user needs. A closer examination of individual responses within Table 16 reinforces the impression that users trust the system's security features. Feedback indicates satisfaction with the data protection and privacy measures, which users view as essential for maintaining confidentiality and preventing unauthorized access. These responses highlight the system's role in enhancing security standards on campus. In summary, the findings from Table 16 validate that the AMA Facial Recognition: Digital Record-logs Management System effectively meets security expectations. The "agree" sentiment among respondents underscores their approval of the system's data protection measures, solidifying its role in providing a secure solution for digital record- logs management and campus safety. Weighted Mean Average: The researchers calculated an overall score for each familiarity group using the weighted mean average and the exact intervals indicated in the survey table. This statistical metric can be used to assign different levels of attention or relevance to comments based on their perceived importance. The formula used for the weighted mean average is: where: Σ (capital sigma) mean "sum". wi is the weight assigned to each value In the dataset. xi is the value in the dataset. Σ(wixi) means the sum of the product of each weight and value. Σ wi means the sum of all weights. Evaluation of the Study The data collection findings for the recommended web-based management system for Steel Fitness Gym are expanded upon in this chapter. The purpose of the test- based poll was to ascertain whether the organisation would benefit from the recommended approach. The researchers assessed the proposed framework's suitability for the business needs after putting it into practice Chapter V Summary, Conclusions, and Recommendations Summary This study provides a comprehensive overview of the research methods used to evaluate the
AMA Facial Recognition: Digital Record-logs Management System proposed for AMA Computer College. The research employed a combination of qualitative and quantitative approaches, including system testing, questionnaires, interviews, and observations. The Likert scale was primarily used to assess the degree of familiarity with and opinions on the facial recognition system. Statistical analysis of the data involved calculating a weighted mean average to assign various weights to familiarity categories, allowing for a nuanced assessment of the system's effectiveness and user acceptance. The study focused on evaluating the system's core functionalities, including facial recognition accuracy, real-time entry/exit logging, data security measures, and overall usability. The research methods were designed to determine how well the system meets the specific needs of AMA Computer College, particularly in enhancing campus security and streamlining administrative processes. Conclusions

**The study's findings provide** crucial **insights into the** effectiveness and practicality **of** **14**

the proposed AMA Facial Recognition: Digital Record-logs Management System. The weighted mean average enabled a thorough assessment of familiarity levels, while the Likert scale facilitated a more indepth analysis of user perceptions and experiences. Key conclusions drawn from the research include: 1. The facial recognition system demonstrates a high level of accuracy in identifying

individuals and logging entry/exit events, significantly enhancing campus security measures. 2. Users generally found the system interface intuitive and easy to navigate, indicating the successful implementation of user-centric design principles. 3. The system's automated logging features show potential for significantly reducing administrative workload and improving the efficiency of record-keeping processes. Overall, the conclusions highlight a positive correlation between the proposed system's capabilities and the organization's requirements, suggesting that the AMA Facial Recognition: Digital Record-logs Management System is wellpositioned to enhance campus security and operational efficiency at AMA Computer College.

**Recommendations Based on the study's findings, the following 12 recommendations are proposed**

for future researchers: 1. System Refinement: Consider refining the facial recognition algorithms and user interface based on feedback received during testing. This could include improving recognition accuracy under varied lighting conditions or enhancing the dashboard to facilitate easier data interpretation. 2. Phased Implementation: Implement the system in phases across the campus, starting with high-traffic areas. This approach allows for gradual adaptation and the identification of any unforeseen challenges in a real-world setting. 3. Continuous User Training: Develop a comprehensive training program for both administrative staff and students to ensure optimal use of the system. Regular workshops and easily accessible online resources can support this initiative. 4. Privacy Policy Development: Collaborate with legal experts to develop and clearly communicate a robust privacy policy that addresses data collection, storage, and usage concerns. 5. Regular System Audits: Establish a schedule for regular system audits to assess performance, security, and compliance with evolving data protection regulations. 6. Feedback Mechanism: Implement a continuous feedback mechanism within the system to capture user suggestions and report any issues, facilitating ongoing improvement. 7. Integration Expansion: Explore possibilities for integrating the facial recognition system with other campus management tools, such as class attendance tracking or library access systems, to maximize its utility. 8. Long-term Performance Monitoring: Set up a long-term monitoring system to track the system's impact on campus security incidents, administrative efficiency, and overall user satisfaction over time. 9. Ethical Use Guidelines: Develop clear guidelines for the ethical use of facial recognition technology on campus, ensuring transparency and building trust within the college community. 10. Scalability Planning: Plan for potential system scalability to accommodate future growth in the student population or expansion to other AMA Computer College campuses. By implementing these recommendations, AMA Computer College can maximize the benefits of the Facial Recognition: Digital Record-logs Management System, while addressing potential challenges and ensuring the long-term success of the project. 37


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